

Tight instances of the Lonely Runner Conjecture: complete classification of one-entry modifications, a new infinite family, and the growth bound

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Abstract

For a set V of $n-1$ distinct positive integers let $\text{LR}(V) = \max_t \min_{v \in V} \|vt\|$, where $\|x\|$ is the distance from x to the nearest integer; V is *tight* if $\text{LR}(V) = 1/n$, the value predicted by the Lonely Runner Conjecture. Goddyn and Wong (*Integers* **6** (2006), #A38) characterised the tight sets obtained from the baseline $[n-1] = \{1, \dots, n-1\}$ by scaling one entry by an integer factor; the characterisation of arbitrary one-entry modifications $[n-1]_{r \rightarrow w}$ has been open since, and the survey of Perarnau and Serra lists it as their first open problem.

We prove four results. **(1) Complete classification** (Theorem 1.2): $[n-1]_{r \rightarrow w}$ with $w > n-1$ is tight if and only if either $2r > n-1$, $r \mid w$, and the Goddyn–Wong criterion holds, or $(n, r, w) \in \{(5, 2, 7), (6, 2, 9)\}$; in particular no tight one-entry modification exists in the mid-range $3 \leq r \leq (n-1)/2$. The proof for $r \geq 3$ is purely theoretical: a single inequality, proved via the Jacobsthal function (Kanold’s bound), eliminates the computational residue of the effective-bound approach; the cases $r = 1$ (Goddyn–Wong theorem) and $r = 2$ (an exact check at five values of n) are elementary. **(2) Multi-swaps** (Theorems 5.1 and 5.2): we recall the under-appreciated multi-acceleration theorem of Goddyn and Wong (simultaneous acceleration of several runners is tight whenever each individual acceleration satisfies their gcd criterion) and isolate an explicit CRT doubling subfamily of it, verifying its members $n = 74$ (their example) and $n = 284$ in exact arithmetic. **(3) Growth bound** (Theorems 1.3 and 6.2): every tight one-entry modification with $n \geq 4$ runners satisfies $\max V \leq 0.60n \log n + 52n$, the constant $\frac{1}{2}$ is sharp along $n = p\# + 2$, and the same bound extends to the Goddyn–Wong multi-acceleration families; whether it extends to arbitrary multi-swaps is open (Question 9.2). **(4) Two-swap classification** (Theorems 7.5, 7.10, 7.13): an isolation lemma shows that no r' -constraint bites on the two components of $U(n, 2)$ around $\frac{1}{2}$, and a coverage-density lemma shows that two bad-interval families, each of density $1/n$, cannot cover them; the exclusion of two-swaps removing $r = 2$ with a mid-range speed is *theoretical* for $n \geq 19$ (odd) and $n \geq 20$ (even). Run at $\frac{1}{q}$ instead of $\frac{1}{2}$, the same mechanism yields the strongest theoretical result of this paper: two-swaps containing *any* removal $2 \leq q \leq n/10$ are excluded for $n \geq \max(40, 10q)$, with no restriction whatsoever on the second removal and none on the speeds (Theorem 7.10). An exact-arithmetic census over all $n \leq 140$, $w \leq 8n$ and all removal pairs then finds exactly two tight instances:

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$\{1, 4, 5, 6, 7, 11, 13\}$ ($n = 8$, Wills) and the Goddyn–Wong doubling instance at $n = 74$ (Theorem 7.13); the residual $9 \leq n \leq 18$ for $R = \{2, 3\}$ is settled by computation within the window $w \leq 40n$ (Theorem 7.6).

In the language of the Lonely Runner spectrum $\tilde{S}(n-1) = \{\text{LR}(V) : |V| = n-1\}$ studied by Giri and Kravitz, the tight sets are the minimisers of $\tilde{S}(n-1)$; our growth bound is an unconditional effective analogue of their conditional finiteness theorem, the deletion value $\text{LR}([n-1] \setminus \{2\})$ is determined in closed form (Theorem 2.8), and the sandwich interval $[1/n, \text{LR}([n-1] \setminus \{r\})]$ contains the isolated region $(1/n, 1/(n-1))$ between the tight value and the first accumulation point whenever $\text{LR}([n-1] \setminus \{r\}) \geq 1/(n-1)$ (Question 2.10).

Finally, we position the classification against the current frontier. The polynomial lower bound on the maximum loneliness due to Bedert approaches the conjectural value from below and so leaves the equality case untouched, and the recent spectrum counterexamples of Fan and Sun delimit how far the Giri–Kravitz pattern can reach. In the shifted variant, where the runners start at adversarially chosen positions, the baseline $[n-1]$ itself fails from five speeds on (Blanco–Criado–Santos); we isolate a monotonicity lemma for the shifted gap (Lemma 7.14) and formulate the shifted analogue of the classification problem (Question 9.4).

1 Introduction

Let $\|x\|$ be the distance from $x \in \mathbb{R}$ to the nearest integer. For a set V of $k = n-1$ distinct positive integers (*speeds*) put

$$\text{LR}(V) = \max_{t \in [0,1)} \min_{v \in V} \|vt\|.$$

The *Lonely Runner Conjecture* of Wills [26] and Cusick [8] asserts $\text{LR}(V) \geq \frac{1}{n}$; it is proved for $n \leq 7$ runners [4, 9, 6, 7, 2], for $n \leq 10$ by computer-assisted methods [20, 25], and announced, in preprints not yet peer-reviewed, up to $n \leq 13$ [21, 23]; see [19] for a survey. For a set of m speeds the best general lower bound on the maximum loneliness is $\frac{1}{2m} + \frac{1}{m^{5/3+o(1)}}$, proved by Bedert [3] with Riesz products, improving Tao’s $\frac{1}{2m} + \frac{(\log m)^{1-o(1)}}{m^2}$ [24]. We emphasise that the present paper does not address the conjecture itself: it concerns the structure of the *equality* case, where the two problems meet at the baseline $[n-1]$, which is tight for every n .

The spectrum $\tilde{S}(m) = \{\text{LR}(V) : |V| = m\}$, studied by Kravitz [17] and Giri–Kravitz [11], collects all loneliness values for m -speed sets. Giri–Kravitz prove (assuming LRC) that each $\tilde{S}(m)$ is well-ordered with order type $\omega^{m-1} + 1$, so the baseline value $1/(m+1)$ has an immediate successor (a *gap* above it), and that the lower accumulation points of $\tilde{S}(m)$ contain $\tilde{S}(m-1)$ (the accumulation structure is governed by the relative spectra of [15]); in particular $1/m$ is a lower accumulation point of $\tilde{S}(m+1)$. (The spectrum values themselves are not governed by Kravitz’s Loneliness Spectrum Conjecture, which Fan and Sun disproved for four speeds by an infinite family of counterexamples [10].) Conditional on LRC, only finitely many sets achieve the minimum $1/(m+1)$ for each m (their Lemma 4.2, via Minkowski’s second theorem, gives a speed bound of $m^{(5/2)m}$). In the notation of this paper ($n-1$ speeds, LRC bound $1/n$), the tight sets are the minimisers of $\tilde{S}(n-1)$, and $1/(n-1)$ is a lower accumulation point. Theorem 1.3 below provides an *unconditional* growth bound $\max V \leq O(n \log n)$ —an effective analogue of this conditional finiteness that does not assume LRC.

The baseline $[n-1] = \{1, \dots, n-1\}$ satisfies $\text{LR}([n-1]) = \frac{1}{n}$ by Dirichlet’s theorem, and a set with $\text{LR}(V) = \frac{1}{n}$ is called *tight*. Perarnau and Serra list the characterisation of tight instances as Problem 1 of their survey, noting that it “has seen no further progress since the work of Goddyn and Wong” [12]. Goddyn and Wong characterised the tight sets of the form $[n-1]_{r \mapsto mr}$ (one entry scaled by an integer):

Theorem 1.1 ([12, Thm. 2.3]). *For $n \geq 2$, $r \in [n-1]$ and $m \geq 1$, the set $[n-1]_{r \rightarrow mr}$ satisfies $\text{LR} \geq \frac{1}{n}$, with equality iff $n = 2$, or $(n, r, m) = (3, 1, 4)$, or*

$$\gcd(r, b) > 1 \quad \text{for every } b \in \{s, s+1, \dots, ms-1\}, \quad s := n-r. \quad (1)$$

For *arbitrary* replacements $[n-1]_{r \rightarrow w} = ([n-1] \setminus \{r\}) \cup \{w\}$ the picture has remained incomplete. Two sporadic tight sets are classical (identified by Wills [27], after a comment of P. Flor; both are listed in [12]):

$$T_1 = \{1, 3, 4, 7\} = [4]_{2 \rightarrow 7} \ (n=5), \quad T_2 = \{1, 3, 4, 5, 9\} = [5]_{2 \rightarrow 9} \ (n=6),$$

and Goddyn–Wong write that the finiteness they prove for each fixed r “partially explains why there are no further sporadic speed sets resembling T_1 and T_2 ”. The effective bound $w \leq 4rI/(2s-I)$ (derived in Section 2 from the component structure of the uncovered region; I is the minimal binding speed, Section 3) makes that finiteness explicit; the key inequality $4rI/(2s-I) < r(n-2)$, proved for all $r \geq 3$ via the Jacobsthal function (Lemma 3.3), eliminates the computational residue entirely. Here we settle every r , and in addition give a quantitative theory of tight multi-swaps: we recall the multi-acceleration theorem of Goddyn and Wong (Theorem 5.1), isolate an explicit doubling family (Corollary 5.2), and establish a growth bound for tight one-entry modifications.

Theorem 1.2 (Complete classification). *Let $n \geq 4$ and let $w > n-1$ be an integer. Then $[n-1]_{r \rightarrow w}$ is tight if and only if one of the following holds:*

- (i) $2r > n-1$, $r \mid w$, and (1) holds with $m = w/r$;
- (ii) $(n, r, w) = (5, 2, 7)$ or $(6, 2, 9)$.

In particular, for $3 \leq r \leq (n-1)/2$ no tight one-entry modification exists.

The proof for $r \geq 3$ is theoretical, with no exhaustive computation. The effective bound $w \leq 4rI/(2s-I)$, derived in Section 2 from the component structure of U (the longest component has length $(2s-I)/(2rnI)$, and the interval lemma forces $w \leq 2/(n \cdot \text{length})$), reduces the problem to the single inequality $4rI/(2s-I) < r(n-2)$; Lemma 3.3 below proves this for *all* $r \geq 3$ via the Jacobsthal function (Kanold’s bound $j(r) \leq 2^{\omega(r)}$, with a single arithmetic check at $r = 6$), thereby eliminating the residual machine verification of the effective-bound approach entirely. The boundary cases are elementary: $r = 1$ is excluded directly by Theorem 1.1 (see the proof), and $r = 2$ leaves the five values $n \in \{5, 6, 7, 8, 10\}$, checked exactly in Section 8.

Our second result quantifies how large the speeds in a tight one-entry modification can be. Goddyn and Wong showed qualitatively that r “must be quite large compared to m and s ” and solved the associated forward optimisation problem (given (m, s) , minimise r); the inverse extremal question — given the number of runners, how large can a speed be? — does not appear in the literature.

Theorem 1.3 (Extremal speed). *Let $M(n) = \max\{\max V : V = [n-1]_{r \rightarrow w} \text{ tight, } w > n-1\}$, with $M(n) = -\infty$ if no such V exists. Then*

$$M(n) < 0.60 n \log n + 52 n \quad \text{for all } n \geq 3, \quad \text{and} \quad M(n) = \left(\frac{1}{2} + o(1)\right) n \log n$$

along the sequence $n = p\# + 2$, where $p\# = \prod_{\ell \leq p} \ell$ is the primorial: the set $[n-1]_{r \rightarrow w}$ with $r = p\#$ and $w = \frac{q-1}{2}r$, q the smallest prime not dividing r , is tight and has $w = (\frac{1}{2} + o(1)) n \log n$.

Corollary 1.4. *The largest speed of a tight instance with n runners is not $O(n)$: along $n = p\# + 2$ it exceeds $(\frac{1}{2} - \varepsilon)n \log n$. In particular no linear bound $B(n) = Cn$ can confine all tight instances, answering (negatively, in a strong form) the linear-window question.*

Remark 1.5 (Bounds for all sets versus bounds for tight sets). Theorem 1.3 is complementary to the lower bounds for the maximum loneliness that hold for *all* speed sets: Tao [24] proved $\text{LR}(V) \geq \frac{1}{2m} + \frac{(\log m)^{1-o(1)}}{m^2}$ for every set V of m speeds, and Bedert [3] improved this to $\text{LR}(V) \geq \frac{1}{2m} + \frac{1}{m^{5/3+o(1)}}$ by Riesz-product methods. These bounds approach the conjectured value $\frac{1}{m+1}$ from below without reaching it, so they do not restrict the equality case at all, while Theorem 1.3 confines the speeds within the equality case; neither implies the other. In spectrum language, the lower bounds locate the minimum of $\tilde{S}(m)$ in $[\frac{1}{2m} + \frac{1}{m^{5/3+o(1)}}, \frac{1}{m+1}]$, where equality on the right is equivalent to the Lonely Runner Conjecture itself (Dirichlet gives $\min \tilde{S}(m) \leq \frac{1}{m+1}$ always), and Theorem 1.3 bounds the speeds of the one-entry modifications attaining that endpoint. For the *shifted* version of the conjecture, where the runners start at adversarially chosen positions, the corresponding questions have recently taken an opposite turn; see Section 7.4.

The proofs are elementary: Dirichlet’s approximation theorem, exact interval arithmetic, and, for Theorem 1.3, classical Chebyshev bounds. All computational claims are certified in exact rational arithmetic, and every step of the proofs has been audited in exact arithmetic (Section 8); the paper is self-contained except for Theorems 1.1 and 5.1 and the Rosser–Schoenfeld bounds.

2 The uncovered region and its components

Throughout, r, s, n, w are integers with $n = r + s$, $V = [n - 1]_{r \rightarrow w}$, $w \geq n$ (equivalent to $w > n - 1$), and $f_W(t) = \min_{v \in W} \|vt\|$ for a finite set W . Define the *uncovered region*

$$U = U(n, r) = \left\{ t \in [0, 1) : \|it\| > \frac{1}{n} \text{ for all } i \in [n - 1] \setminus \{r\} \right\}.$$

Lemma 2.1 (Dirichlet). *For every t there is $1 \leq i \leq n - 1$ with $\|it\| \leq \frac{1}{n}$.*

Lemma 2.2 (Coverage criterion). *If $V = [n - 1]_{r \rightarrow w}$ is tight, then $\|wt\| \leq \frac{1}{n}$ for every $t \in U$.*

Proof. Tightness gives $f_V(t) \leq \text{LR}(V) = \frac{1}{n}$ for all t . For $t \in U$ every speed of V other than w exceeds $\frac{1}{n}$, so $\|wt\| \leq \frac{1}{n}$. $\square$

Lemma 2.3 (Divisibility). *If $2r > n - 1$ and $[n - 1]_{r \rightarrow w}$ is tight, then $r \mid w$.*

Proof. At $t_0 = 1/r$: every $i \in [n - 1] \setminus \{r\}$ has $r \nmid i$ (the only multiple of r in $[n - 1]$ is r), so $\|it_0\| \geq \frac{1}{r} > \frac{1}{n}$; thus $t_0 \in U$ and Lemma 2.2 gives $\|w/r\| \leq \frac{1}{n} < \frac{1}{r}$. Since $\|w/r\| \in \frac{1}{r}\mathbb{Z}_{\geq 0}$, this forces $r \mid w$. $\square$

From now on we work in the *middle range* $2r \leq n - 1$, i.e. $s \geq r + 1$, and assume $r \geq 2$. For $x \not\equiv 0 \pmod{r}$ put

$$F(x) = \max\{i \leq n - 1 : i \neq r, i \equiv x \pmod{r}\}.$$

Since $\{s, s + 1, \dots, n - 1\}$ consists of r consecutive integers and (in the middle range) contains no multiple of r equal to r itself, every nonzero class has a representative there, whence

$$s \leq F(x) \leq n - 1 = r + s - 1 < 2s. \quad (2)$$

Lemma 2.4 (Components of U). *Let $2 \leq r \leq (n-1)/2$. Then*

$$U = \bigsqcup_{\substack{0 \leq p < r \\ \gcd(p,r)=1}} (L_p \sqcup R_p), \quad \begin{aligned} L_p &= \left(\frac{p}{r} - \frac{s}{rnF(u_p)}, \frac{p}{r} - \frac{1}{2rn} \right), \\ R_p &= \left(\frac{p}{r} + \frac{1}{2rn}, \frac{p}{r} + \frac{s}{rnF(r-u_p)} \right), \end{aligned}$$

where $u_p = p^{-1} \bmod r$, and all these open intervals are nonempty.

Proof. Let $t \in U$. By Lemma 2.1 some $i \leq n-1$ has $\|it\| \leq \frac{1}{n}$; by definition of U that i is r , so $|t - \frac{j}{r}| \leq \frac{1}{rn}$ for some integer j . If $g = \gcd(j, r) > 1$ then $r' = r/g \leq r/2 \leq (n-1)/4 < n-1$, $r' \neq r$, and $\|r't\| \leq r'|t - \frac{j}{r}| \leq \frac{1}{gn} < \frac{1}{n}$, contradicting $t \in U$; hence $j \equiv p \pmod{r}$ with $\gcd(p, r) = 1$.

Fix such a p and write $t = \frac{p}{r} + \delta$, $|\delta| \leq \frac{1}{rn}$. The speeds of $[n-1] \setminus \{r\}$ split into multiples of r and the rest.

Multiples. For $i = kr \leq n-1$, $k \geq 2$ (such k exist since $2r \leq n-1$): $\|it\| = kr|\delta|$, and the constraint $\|it\| > \frac{1}{n}$ reads $|\delta| > \frac{1}{krn}$; the binding case is $k = 2$:

$$|\delta| > \frac{1}{2rn}. \quad (3)$$

Non-multiples. For $r \nmid i$ let $\rho = ip \bmod r \in (0, r)$ and $d = \min(\rho, r - \rho) \geq 1$. Since $|i\delta| \leq \frac{n-1}{rn} < \frac{1}{r} \leq \frac{d}{r}$, we get $\|it\| = \frac{d}{r} - i|\delta|$ when δ has the sign opposing ρ 's side ($\delta < 0$ if $\rho = d$, $\delta > 0$ if $\rho = r - d$), and $\|it\| \geq \frac{d}{r} > \frac{1}{n}$ otherwise. The active constraint is therefore one-sided and reads $|\delta| < \frac{dn-r}{rni}$. On each side, the binding constraint has $d = 1$: indeed, by (2) and cross-multiplication, for $d \geq 2$ and any legal $i \leq n-1$,

$$\frac{dn-r}{rni} \geq \frac{2n-r}{rn(n-1)} = \frac{n+s}{rn(n-1)} > \frac{s}{rnF(x)} \quad \text{for every nonzero class } x,$$

because $F(x)(n+s) \geq s(n+s) > s(n-1)$. The classes with $d = 1$ are $i \equiv u_p$ (then $\rho = ip \equiv 1$: left side) and $i \equiv -u_p$ (right side), and among them the largest legal i , namely $F(u_p)$ resp. $F(r-u_p)$, gives the binding bound $|\delta| < \frac{s}{rnF(\cdot)}$.

Combining with (3) yields exactly $L_p \sqcup R_p$; both are nonempty because $F < 2s$ by (2). Conversely every point of $L_p \sqcup R_p$ satisfies all constraints, so it lies in U . $\square$

Lemma 2.5 (Interval lemma). *For $n \geq 3$, the set $\{t : \|wt\| \leq \frac{1}{n}\}$ is a disjoint union of closed intervals $[\frac{j}{w} - \frac{1}{wn}, \frac{j}{w} + \frac{1}{wn}]$ of length $\frac{2}{wn}$ (on $[0, 1]$ the intervals centred at 0 and 1 appear as the two boundary pieces of a single interval of the circle; the components used below lie in $(0, 1)$, away from both). A connected subset lies in one of them; in particular, if $\|wt\| \leq \frac{1}{n}$ on an interval J then some integer j satisfies $\frac{j}{w} - \frac{1}{wn} \leq \inf J$ and $\sup J \leq \frac{j}{w} + \frac{1}{wn}$.*

Proposition 2.6 (Sandwich bound). *For every $w > n-1$ and every $r \in [n-1]$,*

$$\text{LR}([n-1]_{r \rightarrow w}) \leq \text{LR}([n-1] \setminus \{r\}), \quad \text{LR}([n-1]) = \frac{1}{n} \leq \text{LR}([n-1] \setminus \{r\}),$$

where $[n-1] \setminus \{r\}$ has $n-2$ elements. Both inequalities are unconditional: adding a speed cannot increase LR (the first), and $[n-1] \setminus \{r\} \subseteq [n-1]$ with $\text{LR}([n-1]) = \frac{1}{n}$ (the second). The middle value is known exactly in two regimes: for $r = n-1$ it equals $\text{LR}([n-2]) = \frac{1}{n-1}$, and for $r = 2$ it is given in closed form by Theorem 2.8 below. In the language of [11], $1/(n-1)$ is the value achieved by the baseline $[n-2]$ and (assuming LRC) is a lower accumulation point of $\tilde{S}(n-1)$. The interval $[1/n, \text{LR}([n-1] \setminus \{r\})]$ contains every tight modification value unconditionally (the tight value is $1/n$), and (assuming LRC) every modification value.

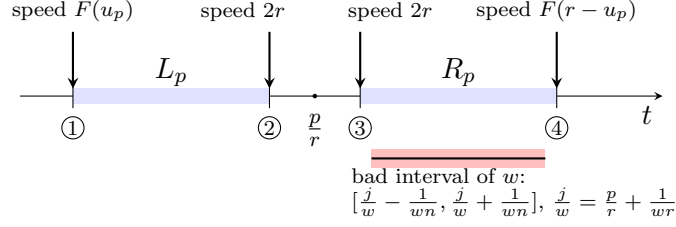


Figure 1: The two components of $U(n, r)$ adjacent to p/r (Lemma 2.4). The edges, left to right, are $\frac{p}{r} - \frac{s}{rnF(u_p)}$, $\frac{p}{r} - \frac{1}{2rn}$, $\frac{p}{r} + \frac{1}{2rn}$, $\frac{p}{r} + \frac{s}{rnF(r-u_p)}$: the inner edges (2), (3) are pinned by the speed $2r$ (constraint (3)), the outer edges (1), (4) by the binding $d = 1$ classes $F(u_p)$, $F(r - u_p)$. For a tight $[n - 1]_{r \rightarrow w}$ the bad intervals of w (bottom, shown at its centre $\frac{j}{w} = \frac{p}{r} + \frac{1}{wr}$, length $\frac{2}{wn}$) must cover every component—the length constraint behind the effective bound $w \leq 4rI/(2s - I)$ and the window lemma.

Proof. Adding a speed cannot increase LR, since $f_{V \cup \{w\}}(t) = \min(f_V(t), \|wt\|) \leq f_V(t)$; applied with $V = [n - 1] \setminus \{r\}$ this gives the first inequality. For the second, LR is decreasing under inclusion of the speed set, so $\text{LR}([n - 1] \setminus \{r\}) \geq \text{LR}([n - 1])$, and $\text{LR}([n - 1]) = \frac{1}{n}$: the value $\frac{1}{n}$ is attained at $t = a/n$ with $\gcd(a, n) = 1$, where $\|ia/n\| = \min(i, n - i)/n \geq \frac{1}{n}$ for all $i \leq n - 1$, while Lemma 2.1 shows that no t has $\|it\| > \frac{1}{n}$ for all $i \leq n - 1$. The same two arguments with $n - 1$ in place of n give $\text{LR}([n - 2]) = \frac{1}{n-1}$, which is the middle value for $r = n - 1$. $\square$

Remark 2.7. The lower bound $\text{LR}([n - 1] \setminus \{r\}) \geq \frac{1}{n-1}$ for all r is the Lonely Runner Conjecture for the family of speed sets $\{1, \dots, n - 1\} \setminus \{r\}$; it is not a consequence of Dirichlet's theorem, which bounds LR from above (Lemma 2.1), and we do not assume it. It is verified in exact arithmetic for all $n \leq 30$, is proved for $r \in \{2, n - 1\}$ (Theorem 2.8 and the proposition above), and holds for explicit further families of r by Proposition 2.9; see Question 2.10.

Theorem 2.8 (Deletion value for $r = 2$). *For every $n \geq 5$,*

$$\text{LR}([n - 1] \setminus \{2\}) = \frac{2}{2\lfloor n/2 \rfloor + 3} > \frac{1}{n - 1}.$$

Proof. Write $D = 2\lfloor n/2 \rfloor + 3$ (so $D = n + 2$ for odd and $D = n + 3$ for even n ; in both cases D is odd, $D - 2 > n - 1$, and $2/D > 1/(n - 1)$), and $V = [n - 1] \setminus \{2\}$.

Lower bound. Let $a = (D + 1)/2$ and $t^* = a/D$. From $2a \equiv 1 \pmod{D}$ we get $\gcd(a, D) = 1$, and the only $i \in \{1, \dots, D - 1\}$ with $ia \equiv \pm 1 \pmod{D}$ are $i = 2$ and $i = D - 2$. Since $D - 2 > n - 1$, every $i \in V$ satisfies $ia \not\equiv 0, \pm 1 \pmod{D}$ (the congruence $ia \equiv 0$ is excluded by coprimality and $i < D$), i.e. $\|it^*\| = |ia \bmod \pm D|/D \geq 2/D$. Hence $f_V(t^*) \geq 2/D$.

Upper bound. Let $t \in [0, 1]$; we show $f_V(t) \leq 2/D$. For $n \in \{5, 6\}$ this is exact computation, so assume $n \geq 7$; then $D \leq 2(n - 2)$, so $\frac{1}{n-2} \leq \frac{2}{D}$. Among the $n - 1$ points $0, t, 2t, \dots, (n - 2)t$ on the unit circle, two cyclically adjacent ones are at distance $\leq \frac{1}{n-1}$, so some $d \in \{1, \dots, n - 2\}$ has $\|dt\| \leq \frac{1}{n-1} \leq \frac{1}{n-2} \leq \frac{2}{D}$. If $d \neq 2$ then $d \in V$ and $f_V(t) \leq \frac{2}{D}$. If $d = 2$, put $e = \|2t\| \leq \frac{1}{n-1}$ and write $2t = m + \epsilon$ with $m = \lfloor 2t \rfloor$, $|\epsilon| = e$. If $e \leq \frac{1}{D}$ then $\|4t\| = 2e \leq \frac{2}{D}$ (and $4 \in V$), where the identity $\|4t\| = 2e$ holds because $2e \leq \frac{2}{n-1} \leq \frac{1}{3} < \frac{1}{2}$. If $e > \frac{1}{D}$ and m is even, then $\|t\| = \frac{e}{2} \leq \frac{1}{2(n-1)} \leq \frac{2}{D}$. Finally, if $e > \frac{1}{D}$ and m is odd, let i_0 be the largest odd element of V : by the definition of D , $i_0 = n - 1$ for even and $i_0 = n - 2$ for odd n , so $i_0 \geq D - 4$ in both cases.

Since $i_0 m$ is odd,

$$\|i_0 t\| = \left\| \frac{1}{2} + \frac{i_0 e}{2} \right\| = \frac{1}{2} - \frac{i_0 e}{2},$$

where the displayed identity uses $\frac{i_0 e}{2} \leq \frac{i_0}{2(n-1)} \leq \frac{1}{2}$ (with equality only for even n). From $e > \frac{1}{D}$ and $i_0 \geq D - 4$,

$$i_0 e > \frac{i_0}{D} \geq \frac{D-4}{D} = 1 - \frac{4}{D}, \quad \text{whence} \quad \|i_0 t\| < \frac{4/D}{2} = \frac{2}{D},$$

and $i_0 \in V$ (note $i_0 \geq 3$), so $f_V(t) \leq \frac{2}{D}$. $\square$

Proposition 2.9 (Witness families). *Let $1 \leq r \leq n-1$ and $V = [n-1] \setminus \{r\}$. In each of the following cases an explicit witness t^* with $f_V(t^*) \geq \frac{1}{n-1}$ is available:*

- (i) $r = n-1$: $t^* = a/(n-1)$ with $\gcd(a, n-1) = 1$;
- (ii) $r \leq n-3$ and $\gcd(r, 2n-3) = 1$: $t^* = a/(2n-3)$ with $a \equiv r^{-1} \pmod{2n-3}$; then $f_V(t^*) \geq \frac{2}{2n-3} > \frac{1}{n-1}$;
- (iii) r odd with $\gcd(r, n-1) = 1$: $t^* = a/(2n-2)$ with $a \equiv r^{-1} \pmod{2n-2}$; then $f_V(t^*) \geq \frac{1}{n-1}$.

Proof. (i) is the computation of $\text{LR}([n-2])$ from the proof of Proposition 2.6. For (ii) and (iii) let $q_0 \in \{2n-3, 2n-2\}$ and let a be the unit of q_0 with $a \equiv r^{-1} \pmod{q_0}$. Suppose $i \in [n-1] \setminus \{r\}$ had $|ia \bmod^{\pm} q_0| \leq 1$. The congruence $ia \equiv 0 \pmod{q_0}$ is impossible ($\gcd(a, q_0) = 1$, $i < q_0$), while $ia \equiv \pm 1$ forces $i \equiv \pm a^{-1} \equiv \pm r \pmod{q_0}$, so $i = r$ or $i = q_0 - r$; the latter is $> n-1$ because $q_0 - r \geq 2n-3 - (n-3) = n$ in (ii) and $q_0 - r = 2n-2 - r \geq n$ in (iii) (as $r \leq n-2$ there). Both are excluded, so $f_V(a/q_0) \geq 2/q_0$, which is $> 1/(n-1)$ in (ii) and $= 1/(n-1)$ in (iii). $\square$

Question 2.10. Is $\text{LR}([n-1] \setminus \{r\}) \geq \frac{1}{n-1}$ for all n and all $r \in [n-1]$? This is the Lonely Runner Conjecture for the speed sets $\{1, \dots, n-1\} \setminus \{r\}$; it holds with equality for $r = n-1$, is proved for $r = 2$ (Theorem 2.8) and for the families of Proposition 2.9 (in particular, for every r coprime to $2n-3$), and is verified in exact arithmetic for all $n \leq 30$. A stronger form: is $\text{LR}([n-1] \setminus \{r\}) = \frac{1}{n-1}$ only for $r = n-1$ (true for $n \leq 30$, and for $r = 2$ by Theorem 2.8)?

3 Proof of Theorem 1.2

Suppose $V = [n-1]_{r \rightarrow w}$ is tight with $3 \leq r \leq (n-1)/2$ and $w \geq n$. By Lemmas 2.2 and 2.4, $\|wt\| \leq \frac{1}{n}$ on the closures $\overline{L_1}$ and $\overline{R_1}$ of the two components adjacent to $t_0 = \frac{1}{r}$ (continuity extends coverage to closures). Write

$$F_+ = F(1), \quad F_- = F(r-1), \quad c \equiv w \pmod{r}, \quad c \in \left(-\frac{r}{2}, \frac{r}{2}\right].$$

Lemma 3.1 (Window lemma). *Assume additionally $w < r(n-2)$. Then:*

- (R) $\|wt\| \leq \frac{1}{n}$ on $\overline{R_1}$ implies $-\frac{w}{2n} - \frac{r}{n} \leq c \leq \frac{r}{n} - \frac{ws}{nF_-}$;
- (L) $\|wt\| \leq \frac{1}{n}$ on $\overline{L_1}$ implies $\frac{ws}{nF_+} - \frac{r}{n} \leq c \leq \frac{w}{2n} + \frac{r}{n}$.

Proof. We prove (R); (L) is symmetric. Put $[a, b] = \overline{R_1}$, so $a = t_0 + \frac{1}{2rn}$, $b = t_0 + \frac{s}{rnF_-}$. By Lemma 2.5 there is $j \in \mathbb{Z}$ with $wb - \frac{1}{n} \leq j \leq wa + \frac{1}{n}$. Write $w = Kr + c$, so $wt_0 = K + \frac{c}{r}$, and put $J = j - K$. Then

$$\frac{c}{r} + \frac{ws}{rnF_-} - \frac{1}{n} \leq J \leq \frac{c}{r} + \frac{w}{2rn} + \frac{1}{n}.$$

If $J \geq 1$: the upper bound gives $c \geq r - \frac{w}{2n} - \frac{r}{n}$, and $w < r(n-2)$ is equivalent to $r - \frac{w}{2n} - \frac{r}{n} > \frac{r}{2}$, contradicting $c \leq \frac{r}{2}$; so $J \geq 1$ is impossible. If $J \leq -1$: the lower bound gives $c \leq -r + \frac{r}{n} - \frac{ws}{nF_-} < -\frac{r}{2}$, impossible since $c > -\frac{r}{2}$. Hence $J = 0$, and the two inequalities displayed are exactly the asserted window for c after multiplying by r . $\square$

Lemma 3.2 (Sign lemma). *If $s \geq r + 1$ and $w \geq n$, then $\frac{r}{n} - \frac{ws}{nF_-} < 0$ and $\frac{ws}{nF_+} - \frac{r}{n} > 0$.*

Proof. Both reduce to $ws > rF_{\pm}$. Since $F_{\pm} \leq n - 1$ and $w \geq n$, it suffices that $ns > r(n - 1)$, i.e. $n(s - r) > -r$, which holds as $s \geq r + 1$. $\square$

Lemma 3.3 (Key inequality). *For all $r \geq 3$ with $2r \leq n - 1$ (i.e. $s \geq r + 1$):*

$$\frac{4rI}{2s - I} < r(n - 2).$$

Proof. Write $d = I - s \geq 0$; the bound is equivalent to $d < \frac{s(n-6)}{n+2}$.

Case A: $n \geq 3r + 1$ (i.e. $s \geq 2r + 1$). Using $d \leq r - 1$ (since $I \leq n - 1 = r + s - 1$): the condition $d < \frac{s(n-6)}{n+2}$ is implied by $(r - 1)(n + 2) < s(n - 6) = (n - r)(n - 6)$, i.e. by

$$g(n) := n^2 - n(2r + 5) + 4r + 2 > 0.$$

The function g is an upward-opening quadratic in n with vertex at $n^* = (2r + 5)/2 < 3r + 1$ (since $r \geq 3$), so g is *increasing* for $n \geq 3r + 1$. Evaluating at the minimum $n = 3r + 1$:

$$g(3r + 1) = (3r + 1)^2 - (3r + 1)(2r + 5) + 4r + 2 = 9r^2 + 6r + 1 - 6r^2 - 17r - 5 + 4r + 2 = 3r^2 - 7r - 2,$$

which is positive for $r \geq 3$ (the positive root of $3x^2 - 7x - 2 = 0$ is $(7 + \sqrt{73})/6 \approx 2.59$). Hence $g(n) > 0$ for all $n \geq 3r + 1$ and $r \geq 3$.

Case B: $n \leq 3r$ (i.e. $s \leq 2r$). Throughout Case B we use that the middle range requires $2r \leq n - 1$, i.e. $n \geq 2r + 1$; combined with $r \geq 3$ this gives $n \geq 7$. Let $a = s \bmod r \in \{0, \dots, r - 1\}$. Recall $d = I - s$, where $I = \min_{\gcd(u, r)=1} \min(F(u), F(r - u))$ and $F(x) = \max\{i \leq n - 1 : i \neq r, i \equiv x \bmod r\}$. Since the block $\{s, \dots, n - 1\}$ has r consecutive integers, $F(x) = s + ((x - s) \bmod r)$ for $x \not\equiv 0 \pmod{r}$, so

$$d = \min_{\gcd(u, r)=1} \min((u - s) \bmod r, (r - u - s) \bmod r).$$

In particular:

B1 If $a = 0$ ($r \mid s$): the class 0 is non-unit, so $d \geq 1$. On the other hand the class 1 is always a unit for $r \geq 3$ and $F(1) = s + 1$, so $d \leq 1$, hence $d = 1$. Now $r \mid s$ and $s \geq r + 1$ force $s \geq 2r$, so $n = r + s \geq 3r \geq 9$ (for $r \geq 3$). Then $\frac{s(n-6)}{n+2} \geq \frac{s(3r-6)}{3r+2}$; since $s \geq 2r$ this is at least $\frac{2r(3r-6)}{3r+2}$, which exceeds 1 for $r \geq 3$ (at $r = 3$ it equals $\frac{6 \cdot 3}{11} = \frac{18}{11} > 1$). Hence $d = 1 < \frac{s(n-6)}{n+2}$.

B2 If $\gcd(a, r) = 1$: then $a \equiv s \pmod{r}$ is a unit, so $F(a) = s + 0 = s$ (the representative $\leq n - 1$ in class a is exactly s), giving $I \leq s$ and $d = 0 < \frac{s(n-6)}{n+2}$ (trivially, since $s \geq 2$ and $n \geq 7$ give $\frac{s(n-6)}{n+2} > 0$).

B3 If $\gcd(a, r) > 1$ and $\gcd(a + 1, r) = 1$: the class $a + 1$ is a unit with $F(a + 1) = s + 1$, so $I \leq s + 1$ and $d \leq 1$. Note that $n = 2r + 1$ gives $a = s \pmod{r} = (r + 1) \pmod{r} = 1$, so $\gcd(a, r) = \gcd(1, r) = 1$: this is Case B2, not B3. Hence B3 requires $n \geq 2r + 2$, giving $s \geq r + 2$. Moreover $a \geq 2$ (since $a = 0$ is B1 and $a = 1$ gives $\gcd(1, r) = 1$, i.e. B2). Thus $s \geq r + a \geq r + 2$ and $n \geq 2r + 2 \geq 8$ (for $r \geq 3$). We verify $d \leq 1 < \frac{s(n-6)}{n+2}$: since $s \geq r + 2$ and $n \geq 2r + 2$, $\frac{s(n-6)}{n+2} \geq \frac{(r+2)(2r-4)}{2r+4} = \frac{(r+2) \cdot 2(r-2)}{2(r+2)} = r - 2 \geq 1$ for $r \geq 3$ (equality at $r = 3$, but B3 does not arise for $r = 3$ since $\{0, 1, 2\}$ contains no a with $\gcd(a, 3) > 1$ and $\gcd(a + 1, 3) = 1$). For $r \geq 4$: $r - 2 \geq 2 > 1$, completing B3.

B4 If $\gcd(a, r) > 1$ and $\gcd(a + 1, r) > 1$: then $d \geq 2$ and $r \geq 6$ (two consecutive non-units require r to have at least two distinct prime factors, so $\omega(r) \geq 2$).

Recall $d = I - s$, $I = \min_{\gcd(u, r)=1} \min(F(u), F(r - u))$ and $F(x) = s + ((x - s) \pmod{r})$. Since the units mod r are symmetric (u is a unit iff $-u$ is), ranging u over units covers both classes u and $-u$, so

$$d = \min_{\gcd(u, r)=1} (u - s) \pmod{r} =: \delta,$$

the distance from $a = s \pmod{r}$ up to the next unit class: the residues $a, a + 1, \dots, a + \delta - 1$ are $\delta \geq d \geq 2$ consecutive non-units mod r . Let $j(r)$ denote the *Jacobsthal function*: the least m such that every m consecutive integers contains one coprime to r . The block $a, \dots, a + \delta - 1$ contains no unit, so $\delta < j(r)$, i.e. $d \leq j(r) - 1$, and Kanold's bound $j(r) \leq 2^{\omega(r)}$ [16] gives $d \leq 2^{\omega(r)} - 1$. We claim $2^{\omega(r)} - 1 < r/2$. Writing $r = p_1^{a_1} \cdots p_k^{a_k}$ with $k = \omega(r) \geq 2$ distinct primes $p_1 < p_2 < \dots$, we have $r \geq p_1 p_2 \cdots p_k$ (the product of the first k primes). We prove $2^k - 1 < (p_1 \cdots p_k)/2$ by induction on $k \geq 2$.

Base $k = 2$: $p_1 p_2 \geq 2 \cdot 3 = 6$ gives equality $3 = 3$ at $r = 6$ (excluded); for $r \geq 10$ (the next integer with $\omega(r) = 2$), $2^2 - 1 = 3 < 5 = 10/2$.

Base $k = 3$: $p_1 p_2 p_3 \geq 2 \cdot 3 \cdot 5 = 30$, and $2^3 - 1 = 7 < 15$.

Inductive step ($k \geq 3$): if $2^k - 1 < r/2$ with $r = p_1 \cdots p_k$, then $r' = r \cdot p_{k+1} \geq 3r$ (since $p_{k+1} \geq p_3 \geq 5 \geq 3$), and

$$\frac{r'}{2} \geq \frac{3r}{2} > \frac{3}{2} \cdot 2(2^k - 1) = 3(2^k - 1) > 2^{k+1} - 1,$$

the last inequality being $3 \cdot 2^k - 3 > 2 \cdot 2^k - 1$, i.e. $2^k > 2$, true for $k \geq 2$. Hence $2^{\omega(r)} - 1 < r/2$ for all $r \geq 7$ with $\omega(r) \geq 2$.

Therefore $d \leq 2^{\omega(r)} - 1 < r/2$, and $\frac{r}{2} < \frac{s(n-6)}{n+2}$ iff $r(n+2) < 2s(n-6)$, i.e. $r(n+2) < 2(n-r)(n-6)$, which holds for $s \geq r + 1$ and $n \geq 2r + 1 \geq 7$: expanding gives $rn + 2r < 2n^2 - 12n - 2rn + 12r$, i.e. $2n^2 - 3(r+4)n + 10r > 0$. This upward-opening quadratic has discriminant $9(r+4)^2 - 80r = 9r^2 - 8r + 144 > 0$ and roots $n_{\pm} = \frac{3(r+4) \pm \sqrt{9r^2 - 8r + 144}}{4}$; for $r \geq 6$ (the B4 range, since $\omega(r) \geq 2$ forces $r \geq 6$) the larger root satisfies $n_+ < 2r + 1$: this is equivalent to $\sqrt{9r^2 - 8r + 144} < 5r - 8$, i.e. $9r^2 - 8r + 144 < 25r^2 - 80r + 64$, i.e.

$16r^2 - 72r - 80 > 0$ (equivalently $2r^2 - 9r - 10 > 0$), which holds for $r \geq 6$ (at $r = 6$ it equals $8 > 0$, and it increases thereafter). Hence both roots lie below $2r + 1$ and the quadratic is positive on $[2r + 1, 3r]$.

For $r = 6$: $j(6) = 4$ (direct: units mod 6 are ± 1 , so the maximal gap is from 1 to 5, length 4), hence $d \leq 3$. The following table lists each n in the range $[2r + 1, 3r] = [13, 18]$ with its case, d , and $\frac{s(n-6)}{n+2}$:

n	s	$a = s \bmod 6$	Case	d
13	7	1	B2	0
14	8	2	B4	≤ 3
15	9	3	B4	≤ 3
16	10	4	B3	≤ 1
17	11	5	B2	0
18	12	0	B1	1

The bound $\frac{s(n-6)}{n+2}$ is increasing in n (for $s = n - 6$ fixed by $r = 6$); at the tightest point $n = 14$ it equals $\frac{8 \cdot 8}{16} = 4 > 3 \geq d$. All other rows satisfy the inequality with larger margin, completing B4.

Remark on the choice of bound. Kanold's $j(r) \leq 2^{\omega(r)}$ is not the sharpest known: Iwaniec [14] proves $j(r) \ll \omega(r)^2 \log^2 \omega(r)$, polynomial in $\omega(r)$. We use Kanold's bound because it is elementary and already sufficient—the requirement here is only $2^{\omega(r)} - 1 < r/2$, i.e. a bound growing slower than r along the sequence of the first $\omega(r)$ primes, and the inductive argument above establishes this directly from the definition, with no sieve machinery. The sharper bounds of Iwaniec would only widen the margin, never reverse the inequality.

□

Proof of Theorem 1.2. The range $2r > n - 1$ is Lemma 2.3 together with Theorem 1.1. For $r = 1$ we have $r \mid w$, so $[n - 1]_{1 \rightarrow w} = [n - 1]_{1 \rightarrow w}$ is of the Goddyn–Wong form with $m = w$: Theorem 1.1 applies, and since $n \geq 4$ excludes $n = 2$ and $(n, r, m) = (3, 1, 4)$, while the criterion (1) cannot hold (it would require $\gcd(1, b) > 1$), the set is not tight.

For $r = 2$, $s = n - 2$, the only unit modulo 2 is $u = 1$, and $F(1)$ is the largest odd number $\leq n - 1$; thus $I = n - 1$ for n even and $I = n - 2$ for n odd. The effective bound gives $w \leq 8$ for n odd and $w \leq 8(n - 1)/(n - 3)$ for n even; with $w \geq n$ this leaves only $n \in \{5, 6, 7, 8, 10\}$, and a direct check of all pairs (n, w) in exact arithmetic shows the only tight sets are $(n, w) = (5, 7)$ and $(6, 9)$, i.e. T_1 and T_2 (the classical sporadic sets of [27]).

Let now $3 \leq r \leq (n - 1)/2$; we show no tight V exists. Put $I = \min_{\gcd(u, r)=1} \min(F(u), F(r - u))$.

By Lemma 3.3, $4rI/(2s - I) < r(n - 2)$. The longest component of U has length $(2s - I)/(2rnI)$, so Lemma 2.5 forces $w \leq 4rI/(2s - I) < r(n - 2)$. Now Lemmas 3.1 and 3.2 apply: coverage of $\overline{R_1}$ forces $c < 0$ while coverage of $\overline{L_1}$ forces $c > 0$ — a contradiction. □

Remark 3.4. The proof fails for $r = 2$ exactly where it should. The window argument requires $w < r(n - 2) = 2n - 4$; for $(n, w) = (5, 7)$ one has $w = 7 > 2n - 4 = 6$ and for $(6, 9)$ one has $w = 9 > 8$, so Lemma 3.1 does not apply — and indeed T_1, T_2 are tight. The two sporadic sets live precisely in the wrap-around regime of the window argument.

4 Proof of Theorem 1.3

By Theorem 1.2, a tight $[n-1]_{r \rightarrow w}$ with $w > n-1$ is either sporadic (speeds ≤ 9) or has $2r > n-1$, $w = mr$ with $m \geq 2$ (as $w > n-1 \geq r$), and (1). In the latter case *every prime in* $[s, ms-1]$ *divides* r : a prime b in that interval has $\gcd(r, b) > 1$, i.e. $b \mid r$.

Proof of the upper bound. Let $\theta(x) = \sum_{\ell \leq x} \log \ell$ (sum over primes) be Chebyshev's function. The primes in $[s, ms-1]$ are distinct divisors of r , so their product is at most $r < n$:

$$\theta(ms-1) - \theta(s-1) \leq \log r < \log n.$$

By Rosser–Schoenfeld [22], $\theta(x) > 0.84x$ for $x \geq 101$ and $\theta(x) < 1.01624x$ for all $x > 0$. If $ms-1 \geq 101$,

$$0.84(ms-1) < 1.01624(s-1) + \log n,$$

whence $ms < 1.1905 \log n + 1.2099(s-1) + 1$. Since $w = mr < mn$ and $s \geq 2$ (for $s=1$ the interval contains $b=1$, coprime to everything, so $m=1$), the right-hand side divided by s is maximised at $s=2$ (the logarithmic coefficient $\frac{1.1905}{s} \log n$ dominates the comparison for every $n \geq 3$):

$$w < n \left(\frac{1.1905}{2} \log n + \frac{1.2099}{2} + \frac{1}{2} \right) < 0.60 n \log n + 1.11 n.$$

If $ms-1 < 101$ then $ms \leq 102$ and $w = mr \leq \frac{102r}{s} < \frac{102n}{s} \leq 51n$ (using $r < n$ and $s \geq 2$). Together with the sporadic speeds ≤ 9 , $M(n) < 0.60 n \log n + 52n$ for all n . Replacing the constants 0.84 and 1.01624 by $1 \mp \varepsilon$ (prime number theorem) gives $M(n) \leq (\frac{1}{2} + o(1)) n \log n$. $\square$

Proof of the lower bound. Let $p\# = \prod_{\ell \leq p} \ell$, $r = p\#$, $n = r+2$ (so $s=2$), and let q be the smallest prime not dividing r , i.e. the successor prime of p . Take $m = \frac{q-1}{2}$ (an integer, q odd, and $m \geq 2$ for $p \geq 3$). Every integer $b \in [2, 2m-1] = [2, q-2]$ has a prime factor $\ell \leq b \leq q-2$; since the primes not exceeding $q-2$ are exactly the primes not exceeding p , we get $\ell \mid r$ and hence $\gcd(r, b) > 1$. Thus (1) holds and $[n-1]_{r \rightarrow mr}$ is tight by Theorem 1.1, with

$$w = \frac{q-1}{2} r = \frac{q-1}{2\theta(p)} \cdot \frac{r}{n} \cdot n\theta(p) \sim \frac{1}{2} n \log n,$$

since $\log r = \theta(p) \sim p$ and $q \sim p$ by the prime number theorem. Moreover $m = \frac{q-1}{2}$ is the largest admissible multiplier for this (r, s) : for $m' = \frac{q+1}{2}$ the interval $[2, 2m'-1] = [2, q]$ contains the prime $q \nmid r$. $\square$

Remark 4.1. The extremal ratio $w/(n \log n)$ approaches $\frac{1}{2}$ from above very slowly. The computation of $M(n)$ for all $n \leq 40000$ (Section 8) is a finite enumeration: since $m \geq 2$ is equivalent to $b_0 \geq 2s$ (where b_0 is the smallest integer $\geq s$ coprime to r), and $b_0 - s \leq j(r) \leq 2^{\omega(r)}$ by Kanold's bound, for $r \leq 40000$ we have $\omega(r) \leq 6$ and hence $s \leq 64$; the remaining r range is $[s+1, 40000] \times \{s \leq 64\}$, a finite set of $\sim 2.6 \cdot 10^6$ pairs, each requiring one gcd scan. The global maximum ratio 0.9246 is attained at $n = 212$ (i.e. $r = 210$, $w = 1050$).

Remark 4.2 (Effective finiteness in the spectrum). Giri–Kravitz [11] prove, conditional on LRC, that only finitely many m -speed sets achieve $\min \tilde{S}(m) = 1/(m+1)$ (their Lemma 4.2, via Minkowski's second theorem, gives a speed bound of $m^{(5/2)m}$). Theorem 1.3 provides an *unconditional* and much sharper analogue: the growth bound $\max V \leq O(n \log n)$ depends on no conjecture and is polynomial rather than exponential. The gap between $n \log n$ and $m^{(5/2)m}$ is

not merely technical: the Minkowski bound applies to *all* tight sets, while our bound exploits the one-entry structure through the Goddyn–Wong criterion, so the two are complementary rather than competing. In spectrum language, the tight one-entry modifications are the minimisers of $\tilde{S}(n-1)$, and Theorem 1.3 confines them to speeds $O(n \log n)$ without assuming LRC. A further point of contact is the sandwich interval of Proposition 2.6: the well-ordering of $\tilde{S}(n-1)$ (order type $\omega^{n-2} + 1$, conditional on LRC) in particular isolates the minimum from its successor, and the interval $(1/n, 1/(n-1))$ is exactly the region where that isolation is visible for modifications of the baseline; the transition observed in Section 8.2—the smallest non-tight modification value jumping past $1/(n-1)$ at $n = 9$ —exhibits, on the class of modifications, the boundary of the isolated region predicted by the spectral structure.

5 Tight multi-swaps: the Goddyn–Wong multi-acceleration theorem

The sufficiency direction of the Goddyn–Wong criterion (Theorem 1.1) was proved by them *via* a considerably more general statement, which does not seem to be as widely known as it deserves and which we could not find stated outside [12, Section 3]: several runners may be accelerated simultaneously, each by its own multiplier, and the per-runner Goddyn–Wong criterion alone already guarantees tightness.

Theorem 5.1 ([12, Thm. 3.1]). *Let $m_1, \dots, m_{n-1}$ be positive integers and $[n-1]_{\mathbf{m}} := \{m_1, 2m_2, \dots, (n-1)m_{n-1}\}$. If*

$$\gcd(r, b) > 1 \quad \text{for every } r \in [n-1] \text{ with } m_r \geq 2 \text{ and every } b \in \{n-r, \dots, m_r(n-r)-1\}, \quad (4)$$

then $\text{LR}([n-1]_{\mathbf{m}}) = \frac{1}{n}$. (The condition is vacuous for non-accelerated r , i.e. $m_r = 1$.)

Goddyn and Wong illustrate Theorem 5.1 with the instance obtained from $\{1, \dots, 73\}$ by doubling *both* 70 and 72, i.e. $V = \{1, \dots, 73\} \setminus \{70, 72\} \cup \{140, 144\}$ with $n = 74$, each single doubling being tight by Theorem 1.1; the example is also quoted by Tao [24]. Their Section 3 closes by constructing tight sets in which arbitrarily many runners are properly accelerated, and the survey of Perarnau and Serra accordingly regards the families “obtained by appropriately accelerating runners” as the known source of tight instances, asking for *other* infinite families [19]. The two-swap census of Section 7 gives this dichotomy a quantitative form at small n : up to $n \leq 140$ and $w \leq 8n$, the only tight two-swaps are the sporadic $\{1, 4, 5, 6, 7, 11, 13\}$ of Wills [27] and the Goddyn–Wong instance at $n = 74$ (Theorem 7.13).

For the census, and as a simple explicit supply of tight two-swaps and three-swaps inside Theorem 5.1, we record the following CRT subfamily with all multipliers equal to 2 (for $k = 1$ it is a special case of Theorem 1.1 with $m = 2$).

Corollary 5.2 (Doubling family). *For $1 \leq k \leq 3$, let $s_i = 2^i$ and $P_i = \prod \{p \text{ prime} : s_i \leq p \leq 2s_i - 1\}$ ($i = 1, \dots, k$). The P_i are pairwise coprime. For any n satisfying*

$$n \equiv s_i \pmod{P_i}, \quad i = 1, \dots, k,$$

(infinitely many such n exist by the Chinese Remainder Theorem), the set

$$V = [n-1] \setminus \{n-s_1, \dots, n-s_k\} \cup \{2(n-s_1), \dots, 2(n-s_k)\}$$

is tight: $\text{LR}(V) = 1/n$.

Proof. Put $r_i = n - s_i$; we verify (4) with $m_{r_i} = 2$ for $i \leq k$ and $m_r = 1$ for all other $r \in [n - 1]$, and then apply Theorem 5.1. First $r_i \in [n - 1]$, i.e. $n > s_i$: the CRT conditions force $n \equiv 2 \pmod{6}$; for $k = 1$ the solutions are $n \equiv 2 \pmod{6}$ and $n = 2$ is degenerate ($r_1 = 0 \notin [n - 1]$), so $n \geq 8 > 2 = s_1$; for $k = 2$ every solution satisfies $n \equiv 74 \pmod{210}$, so $n \geq 74 > s_2 = 4$; for $k = 3$ every solution satisfies $n \equiv 23174 \pmod{30030}$, so $n \geq 23174 > s_3 = 8$. Now fix $i \leq k$ and $b \in [s_i, 2s_i - 1] = [s_i, m_{r_i}(n - r_i) - 1]$. If b is prime then $b \mid P_i$, hence $b \mid (n - s_i) = r_i$ by the CRT condition, so $\gcd(r_i, b) = b > 1$. If b is composite then (for $i \leq 3$; Remark 5.3) b is divisible by 2 or by 3; since $P_1 = 6$ forces $n \equiv 2 \pmod{6}$, each $r_i = n - 2^i$ is even, and $r_3 = n - 8 \equiv 2 - 8 \equiv 0 \pmod{3}$; the two composites in $[8, 15]$ not divisible by 2, namely 9 and 15, use $3 \mid r_3$. In all cases $\gcd(r_i, b) > 1$, as required. $\square$

Remark 5.3. The restriction $k \leq 3$ is essential for this CRT pattern: for $k \geq 4$ condition (4) fails for composite $b \in [s_i, 2s_i - 1]$ whose prime factors all lie below s_i . For instance at $k = 4$, $i = 4$ ($s_4 = 16$), $b = 21 = 3 \cdot 7$: the CRT conditions force $r_4 \equiv 1 \pmod{3}$ and $r_4 \equiv 2 \pmod{7}$, so $\gcd(r_4, 21) = 1$. For $k \leq 3$ every composite $b \in [s_i, 2s_i - 1]$ is divisible by 2 or 3 (there are none for $i = 1$; for $i = 2$ they are 4, 6; for $i = 3$ they are 8, 9, 10, 12, 14, 15). This is a limitation of the *pattern*, not of Theorem 5.1, which admits arbitrarily many accelerated runners with general multipliers [12, Cor. 3.3].

The first two members of the doubling family beyond $k = 1$ are $n = 74$ (the Goddyn–Wong example above) and $n = 284$; both are verified in exact rational arithmetic in Section 8 ($n = 284$ by branch-and-bound, 60,979 cells), as a check on the census. The growth is $\max V = 2(n - 2) = O(n)$, well within the $O(n \log n)$ bound of Theorem 1.3.

Remark 5.4. The intervals $[2^i, 2^{i+1} - 1]$ are pairwise disjoint, so the prime sets are disjoint, making P_i pairwise coprime. The CRT solution exists and is unique modulo $\prod P_i$, which grows super-exponentially in k (by Bertrand’s postulate, $P_i \geq 2$ for each i).

6 The growth bound for tight instances

Theorem 1.3 bounds $\max V$ for tight *one-entry* modifications. We now extend the bound to multi-swap modifications $V = ([n - 1] \setminus R) \cup W$ under a per-speed Goddyn–Wong hypothesis (Theorem 6.2) that the k -GW family satisfies, and we identify two obstructions that prevent tightness alone from supplying the hypothesis in general (Question 9.2). The key tool is an extension of the divisibility lemma (Lemma 2.3) from the range $r > (n - 1)/2$ to *all* $r < n$.

Lemma 6.1 (Extended divisibility). *Let V be tight with n runners, and let $r \in [n - 1] \setminus V$ with $r < n$. Then the speed $v \in V$ covering $t_0 = 1/r$ satisfies $r \mid v$.*

Proof. Since $\text{LR}(V) \leq 1/n$ (tightness), for $t_0 = 1/r$ there exists $v \in V$ with $\|v/r\| \leq 1/n$. Now $\|v/r\| = \min(v \bmod r, r - v \bmod r)/r$, so the condition reads $\min(v \bmod r, r - v \bmod r) \leq r/n$. Since $r < n$, we have $r/n < 1$, so $\min(v \bmod r, r - v \bmod r) = 0$, i.e. $r \mid v$. $\square$

This extends Lemma 2.3 (which required $2r > n - 1$) to all $r < n$: the condition $r/n < 1$ suffices.

Theorem 6.2 (Growth bound). *For every tight one-entry modification $V = [n - 1]_{r \rightarrow w}$ with $n \geq 4$ runners, $\max V \leq 0.60 n \log n + 52n$, and the constant $\frac{1}{2}$ is sharp in the sense of Theorem 1.3 ($M(n) = (\frac{1}{2} + o(1)) n \log n$ along $n = p\# + 2$). More generally, let $V = ([n - 1] \setminus R) \cup W$ be*

a tight multi-swap modification such that every $w \in W$ is a multiple $w = mr$ of some $r \in R$ ($m \geq 2$) for which the Goddyn–Wong criterion (1) holds; then $\max V \leq 0.60 n \log n + 52 n$. Every application of the multi-acceleration Theorem 5.1 satisfies this hypothesis; in particular the doubling family of Corollary 5.2 has $\max V = 2(n - 2) = O(n)$.

Proof. The one-entry case is Theorem 1.3: the classification (Theorem 1.2) reduces it to the GW regime, where (1) holds, and the sharp constant comes from the primordial family, whose members are one-entry modifications.

For the multi-swap statement, let $w \in W$. By hypothesis $w = mr$ with $r \in R$ and (1) for the pair (r, m) , where $s = n - r \geq 1$. Every prime $b \in [s, ms - 1]$ has $\gcd(r, b) > 1$, hence $b \mid r$, so exactly as in the proof of Theorem 1.3,

$$\theta(ms - 1) - \theta(s - 1) \leq \log r < \log n,$$

and the Chebyshev estimates of Section 4 give $w = mr \leq 0.60 n \log n + 52 n$. Retained speeds are at most $n - 1$, so $\max V \leq 0.60 n \log n + 52 n$. For the doubling family, the proof of Corollary 5.2 verifies (4), which is (1) with $m = 2$ for every pair $(r_i, 2r_i)$, so the hypothesis holds and $\max V = 2(n - 2)$. $\square$

Remark 6.3. The hypothesis of Theorem 6.2 is automatic for one-entry modifications (Theorem 1.2) and for every application of the multi-acceleration Theorem 5.1, so both bounds are unconditional. For general tight multi-swaps the hypothesis can fail in two ways (Question 9.2). First, if R contains a mid-range $r \leq (n - 1)/2$, the covering speed of Lemma 6.1 may be a *retained* speed: multiples of r other than r itself survive in $[n - 1] \setminus R$, so divisibility constrains no element of W . This occurs in the tight set $\{1, 4, 5, 6, 7, 11, 13\}$ ($n = 8$), where $r = 2$ is covered by the retained speed 4, not by 11 or $13 \in W$. Second, even in the GW regime ($r > (n - 1)/2$) with $k \geq 2$, the tightness of V does not imply that $w = mr$ satisfies (1): other elements of W may cover the bad points of the single-swap. In both cases the condition on W is a *union* condition: components of the joint uncovered region $U(n, R) \supseteq U(n, r)$ (removing more speeds weakens fewer constraints) may be covered piecemeal by several speeds, so the interval-lemma bound $w \leq 2/(n|J|)$ does not apply per-element. The splitting is real: at $n = 8$, $R = \{2, 3\}$, the speed $w = 13$ covers only a sub-interval of a component of $U(8, \{2, 3\})$, and the per-component bound $w \leq 11.2$ (from $U(8, 2)$) is violated by $w = 13$. No counterexample to the $O(n \log n)$ bound was found in the exact-arithmetic searches of Section 8 ($k \leq 3$, $n \leq 140$; $k = 4$, $n \leq 30$); the largest speeds in known tight multi-swaps are $2(n - 2)$ (doubling family) and 13 (the $n = 8$ sporadic), both $O(n)$.

7 Two-swap classification

The correct tightness condition for a two-swap $V = ([n - 1] \setminus \{r_1, r_2\}) \cup \{w_1, w_2\}$ is: for every $t \in U(n, \{r_1, r_2\})$, $\min(\|w_1 t\|, \|w_2 t\|) \leq 1/n$. This is a *union condition*: the bad intervals of w_1 and w_2 together must cover $U(n, \{r_1, r_2\})$. Coverage can split *within* a component (different sub-intervals covered by different w), which defeats the direct generalisation of the window argument of Section 3.

For two-swaps that remove the speed 2 together with a mid-range speed, the split phenomenon is nevertheless excluded *in the large* by a measure-theoretic obstruction: no r' -constraint bites on the two components of $U(n, 2)$ around $t = \frac{1}{2}$, and the bad sets of two speeds, each of density $1/n$, are too sparse to cover them. This yields an unconditional classification for odd

$n \geq 19$ and even $n \geq 20$ (Theorem 7.5) and, for $R = \{2, 3\}$, the full classification (Theorem 7.6) up to a finite speed window for $9 \leq n \leq 18$. The same mechanism, run at $\frac{1}{q}$ instead of $\frac{1}{2}$, in fact excludes two-swaps containing *any* removal q with $2 \leq q \leq n/10$, with no restriction whatsoever on the second removal r' and none on the speeds (Theorem 7.10); the residual open regime is that of two *large* removals (Question 9.1).

7.1 The case $R = \{2, r'\}$: a theoretical classification

Throughout this subsection $R = \{2, r'\}$ with $3 \leq r' \leq (n-1)/2$ (both removed speeds mid-range), $n \geq 8$, and $V = ([n-1] \setminus R) \cup \{w_1, w_2\}$ with $w_1 < w_2$, $w_i \geq n$.

Lemma 7.1 (The components of $U(n, 2)$). *Let $n \geq 5$ and let $F(1)$ be the largest odd integer $\leq n-1$ (so $F(1) = n-2$ for n odd and $F(1) = n-1$ for n even). Then*

$$U(n, 2) = L \sqcup R, \quad L = \left(\frac{1}{2} - \beta, \frac{1}{2} - \frac{1}{4n}\right), \quad R = \left(\frac{1}{2} + \frac{1}{4n}, \frac{1}{2} + \beta\right),$$

with $\beta = \frac{1}{2n}$ for n odd and $\beta = \frac{n-2}{2n(n-1)}$ for n even. Both components have length

$$\ell := \beta - \frac{1}{4n} = \begin{cases} \frac{1}{4n} & (n \text{ odd}), \\ \frac{n-3}{4n(n-1)} & (n \text{ even}). \end{cases}$$

Proof. Specialise Lemma 2.4 to $r = 2$ ($2 \cdot 2 \leq n-1$ for $n \geq 5$): the only unit is $p = 1$, $u_1 = 1$, and $F(1) = F(r - u_1)$ is as stated. (The closed form is re-verified against the exact interval computation by `r2_density.py`.) $\square$

Lemma 7.2 (Isolation). *For every mid-range $r' \geq 3$ and every $t \in L \cup R$ we have $\|r't\| > 1/n$. Consequently $L \cup R \subseteq U(n, \{2, r'\})$, and if V is tight then the bad sets of w_1 and w_2 cover all of $L \cup R$.*

Proof. Write $t = \frac{1}{2} + \tau$ with $\frac{1}{4n} < |\tau| \leq \beta \leq \frac{1}{2n}$. Every point of $L \cup R$ lies in $U(n, 2)$, hence satisfies $\|it\| > 1/n$ for all $i \in [n-1] \setminus \{2\}$; it remains to check $i = r'$, and the 2-constraint is void since $2 \in R$.

If r' is odd, then $r't = \frac{r'}{2} + r'\tau$ with $|r'\tau| \leq r'/(2n) < \frac{1}{2}$, so

$$\|r't\| = \left\|\frac{1}{2} + r'\tau\right\| = \frac{1}{2} - |r'\tau| \geq \frac{1}{2} - \frac{r'}{2n} > \frac{1}{n},$$

where the last inequality is $n > r' + 2$, and this holds because $n \geq 2r' + 1 \geq r' + 4$ for $r' \geq 3$. If r' is even, then $\frac{r'}{2} \in \mathbb{Z}$ and $\|r't\| = \|r'\tau\| = r'|\tau| > r' \cdot \frac{1}{4n} \geq \frac{1}{n}$: strictly, because $|\tau| > \frac{1}{4n}$ is strict for $r' = 4$, and because $r'/4 > 1$ for $r' \geq 6$.

Thus $L \cup R \subseteq U(n, \{2, r'\})$. Tightness gives $f_V(t) \leq 1/n$ for all t , and on $U(n, \{2, r'\})$ every retained speed exceeds $1/n$, so $\min(\|w_1 t\|, \|w_2 t\|) \leq 1/n$ on $L \cup R$: the bad sets B_{w_1}, B_{w_2} cover it. $\square$

Lemma 7.3 (Coverage density). *Let $w \geq n$ and $B_w = \{t \in [0, 1) : \|wt\| \leq 1/n\}$. For every interval $J \subseteq [0, 1)$,*

$$|B_w \cap J| \leq \left(w|J| + \frac{2}{n} + 1\right) \cdot \frac{2}{wn}.$$

Proof. B_w is the union of the $w + 1$ closed intervals $[\frac{j}{w} - \frac{1}{wn}, \frac{j}{w} + \frac{1}{wn}]$, $0 \leq j \leq w$ (modulo 1). Those meeting J have $\frac{j}{w} \in J \pm \frac{1}{wn}$, so j ranges over an interval of length $w|J| + \frac{2}{n}$, containing at most $w|J| + \frac{2}{n} + 1$ integers; each contributes at most $\frac{2}{wn}$ to $B_w \cap J$. $\square$

Lemma 7.4 (Edge reach). *Let $w \geq n \geq 5$. If some bad interval of w covers points of R arbitrarily close to $\inf R = \frac{1}{2} + \frac{1}{4n}$, then $w \geq 2n - 4$.*

Proof. Let $[\frac{j}{w} - \frac{1}{wn}, \frac{j}{w} + \frac{1}{wn}]$ be such an interval; its centre c lies in $(\inf R - \frac{1}{wn}, \inf R + \frac{1}{wn}]$. Since $w \geq n \geq 5$,

$$\inf R - \frac{1}{wn} \geq \frac{1}{2} + \frac{1}{4n} - \frac{1}{5n} > \frac{1}{2},$$

so $c = \frac{1}{2} + \frac{k}{2w}$ with $k = 2j - w \geq 1$ an integer. Then

$$\frac{1}{2w} \leq \frac{k}{2w} = c - \frac{1}{2} \leq \frac{1}{4n} + \frac{1}{wn},$$

which is equivalent to $2n \leq w + 4$. $\square$

Theorem 7.5 (Two-swap with $r = 2$: large- n exclusion). *Let $R = \{2, r'\}$ with $3 \leq r' \leq (n-1)/2$ (both mid-range). No tight two-swap $([n-1] \setminus R) \cup \{w_1, w_2\}$ with $w_i \geq n$ exists for odd $n \geq 19$ or even $n \geq 20$.*

Proof. Suppose V tight. By Lemma 7.2, $L \cup R \subseteq B_{w_1} \cup B_{w_2}$. The points of R arbitrarily close to $\inf R$ are covered by finitely many closed bad intervals, so one interval of one of the speeds, w_* , covers them all; Lemma 7.4 gives $w_* \geq 2(n-2)$, and the other speed is $\geq n$. By Lemma 7.3 the part of B_w inside each of the two components (each of length ℓ) is at most $\beta_w := (w\ell + \frac{2}{n} + 1)\frac{2}{wn}$, and β_w decreases in w , so the total coverable measure is at most

$$2(\beta_{2(n-2)} + \beta_n) = 2\left[\left(2(n-2)\ell + \frac{2}{n} + 1\right)\frac{1}{(n-2)n} + \left(n\ell + \frac{2}{n} + 1\right)\frac{2}{n^2}\right]. \quad (5)$$

If (5) is $< 2\ell = |L \cup R|$, coverage is impossible—a contradiction.

For odd n ($\ell = \frac{1}{4n}$), (5) $< 2\ell$ is an explicit rational inequality in n ; it holds at $n = 19$ and $n = 21$, and for $n \geq 23$ we use the cruder estimate $\beta_w \leq \frac{2\ell}{n} + \frac{2.2}{n^2}$ (valid for $n \geq 20$, $w \geq n$), with which (5) is at most $4 \cdot (\frac{2\ell}{n} + \frac{2.2}{n^2}) = \frac{8\ell}{n} + \frac{8.8}{n^2}$, and $\frac{8\ell}{n} + \frac{8.8}{n^2} < 2\ell$ is equivalent to $\frac{4}{n} + \frac{17.6}{n^2} < 1$, i.e. $n \geq 23$. For even n ($\ell = \frac{n-3}{4n(n-1)}$) the inequality holds at $n = 20$ and $n = 22$, and for $n \geq 24$ the same estimate reduces it to

$$\frac{4}{n} + \frac{17.6(n-1)}{n(n-3)} < 1,$$

whose left side decreases in n and equals 0.9699 at $n = 24$. (The four boundary instances $n \in \{19, 21, 20, 22\}$ are verified in exact rational arithmetic by `r2_density.py`, which verifies the inequality directly for every $19 \leq n \leq 199$, re-derives Lemmas 7.1 and 7.2, and audits Lemma 7.4 by brute force over all intervals.) $\square$

Theorem 7.6 ($R = \{2, 3\}$ two-swap classification). *Within the speed window $w_1, w_2 \leq 40n$, the two-swap $([n-1] \setminus \{2, 3\}) \cup \{w_1, w_2\}$ with $w_i \geq n$ is tight if and only if $n = 8$ and $\{w_1, w_2\} = \{11, 13\}$; the exclusion of $9 \leq n \leq 18$ is computational (exact arithmetic, Section 8), while for odd $n \geq 19$ and even $n \geq 20$ it is unconditional in the speeds (Theorem 7.5, applicable since $r' = 3$ is mid-range for $n \geq 7$).*

Proof. The $n = 8$ instance is certified exactly: $\text{LR} = 1/8$ on the breakpoint set $T(V)$, rediscovered independently by the census and the union-filter search (Section 8). For $9 \leq n \leq 18$, the census enumerates every pair $n \leq w_1 < w_2 \leq 40n$: candidate speeds are constrained by the edge-reach conditions of Lemma 7.4 type (an interval covering the left end a of a component has $j \in [wa - \frac{1}{n}, wa + \frac{1}{n}]$, at most two integers per speed), surviving pairs are screened in floats, and every survivor is re-verified in exact rational arithmetic against the exact components of $U(n, \{2, 3\})$: no pair covers. For odd $n \geq 19$ and even $n \geq 20$, Theorem 7.5 applies. $\square$

Remark 7.7 (Why density beats windows here). Each speed's bad set has density exactly $1/n$ on large scales, so two speeds cover at most a $\frac{4}{n}$ -fraction of any region, up to the per-component slack of Lemma 7.3. Since $|L \cup R|$ is a fixed $\approx \frac{1}{2n}$ and no r' -constraint can shrink it (Lemma 7.2), coverage is impossible once $\frac{4}{n}$ plus slack drops below 1. The argument is insensitive to the split phenomenon—it does not care *which* speed covers *which* part, within or across components—and requires no upper bound on the speeds. Run at $\frac{1}{q}$ rather than $\frac{1}{2}$, the mechanism needs a guaranteed window of measure $\gg \frac{1}{n^2}$ adjacent to $\frac{1}{q}$, which the class structure modulo q supplies as long as $q \leq n/10$ (Section 7.2). What it cannot handle is two removals both close to $\frac{n}{2}$: the guaranteed region $U(n, r_1) \cup U(n, r_2) \subseteq U(n, \{r_1, r_2\})$ can have measure as small as $O(\frac{n-2r}{n^2})$ (when $F(x) \equiv n-1$), which tends to 0 as $r \rightarrow \frac{n-1}{2}$, while the slack accumulates over its $\asymp \varphi(r)$ components. That regime is settled computationally (Theorem 7.13) and remains open in general (Question 9.1).

7.2 Arbitrary small removals

The mechanism of Section 7.1 is special to the removed speed 2 only in two cosmetic respects: the guaranteed window sits at $\frac{1}{2}$, and the classes pinching it are the odd speeds. We now run the same mechanism at $\frac{1}{q}$ for an arbitrary removed speed q . The price is a smallness hypothesis on q ; the gain is substantial: the second removal r' becomes completely unrestricted (mid-range or not), and the speeds remain unbounded. Throughout this subsection,

$$q \geq 2, \quad 4q \leq n-1, \quad R = \{q, r'\}, \quad r' \in [n-1] \setminus \{q\}, \quad S = [n-1] \setminus R,$$

and we define

$$F_1 = \max\{i \in S : i \equiv 1 \pmod{q}\}, \quad F_{-1} = \max\{i \in S : i \equiv -1 \pmod{q}\}, \quad m_0 q = \min\{i \in S : q \mid i\}.$$

The block $\{n-q, \dots, n-1\}$ of q consecutive integers meets every residue class, so removing the single speed r' can push $F_{\pm 1}$ down by at most one block: $n-2q \leq F_{\pm 1} \leq n-1$. Likewise m_0 is 2 or 3 (at most one of $2q, 3q$ is removed, and $3q \leq n-1$), so $m_0 q < \frac{3}{4}n < n$. Two crude consequences of $4q \leq n-1$ drive everything below:

$$F_{-1} \geq n-2q \geq \frac{n+1}{2} > \frac{n-1}{2}, \quad \beta := \frac{1/q - 1/n}{F_{-1}} \leq \frac{n-q}{qn(n-2q)} < \frac{2}{q(n+1)} < \frac{1}{n}. \quad (6)$$

Lemma 7.8 (Windows at $1/q$). *In the setting above, the open intervals*

$$W_- = \left(\frac{1}{q} - \frac{1/q - 1/n}{F_1}, \frac{1}{q} - \frac{1}{m_0 q n} \right), \quad W_+ = \left(\frac{1}{q} + \frac{1}{m_0 q n}, \frac{1}{q} + \frac{1/q - 1/n}{F_{-1}} \right)$$

are nonempty and contained in $U(n, R)$. Consequently, if $V = ([n-1] \setminus R) \cup \{w_1, w_2\}$ with $w_i \geq n$ is tight, then the bad sets of w_1, w_2 cover $W_- \cup W_+$.

Proof. We treat W_+ , writing $t = \frac{1}{q} + \tau$ with $\frac{1}{m_0qn} < \tau < \beta$; the case of W_- is identical with the classes 1 and -1 interchanged (equivalently, apply $t \mapsto 1 - t$).

Nonemptiness. $\frac{1}{m_0qn} < \beta$ is equivalent to $F_{-1} < m_0(n - q)$, and indeed $F_{-1} \leq n - 1 < 2(n - q) \leq m_0(n - q)$, where $2(n - q) > n - 1$ because $q \leq \frac{n-1}{4} < \frac{n-1}{2}$.

Multiples of q . Let $k \geq m_0$ with $kq \in S$. The constraint $\|kqt\| > 1/n$, i.e. $\|kq\tau\| > \frac{1}{n}$, fails exactly on the zones $[\frac{j}{kq} - \frac{1}{kqn}, \frac{j}{kq} + \frac{1}{kqn}]$ of τ , $j \geq 0$. For $j = 0$ these zones lie below $\frac{1}{m_0qn}$; for $j \geq 1$ their lower ends are at least $\frac{1}{kq}(1 - \frac{1}{n}) \geq \frac{n-1}{(n-1)n} = \frac{1}{n}$ (here $kq \leq n - 1$), which exceeds β by (6). So no zone meets the window.

The other classes. Let $i \in S$ have residue $\rho = i \bmod q \in \{1, \dots, q-1\}$, so that $it = A + \frac{\rho}{q} + i\tau$ for an integer A . We claim

$$i\tau < \frac{q - \rho}{q} - \frac{1}{n}. \quad (7)$$

For $\rho = q - 1$ we have $i \leq F_{-1}$, hence $i\tau < i \frac{1/q - 1/n}{F_{-1}} \leq \frac{1}{q} - \frac{1}{n}$, which is (7). For $\rho \leq q - 2$, (6) gives $i \leq n - 1 < 2F_{-1}$ and $(q - \rho)n - q \geq 2n - q > 2(n - q)$, whence

$$i\tau < \frac{i(n - q)}{qnF_{-1}} < \frac{2(n - q)}{qn} < \frac{(q - \rho)n - q}{qn} = \frac{q - \rho}{q} - \frac{1}{n}.$$

Given (7), the point $z := \frac{\rho}{q} + i\tau$ lies in $(\frac{\rho}{q}, 1 - \frac{1}{n})$, so $\|it\| = \|z\| > \frac{1}{n}$: if $z \leq \frac{1}{2}$ then $\|z\| = \min(z, 1 - z) = z \geq \frac{\rho}{q} \geq \frac{1}{q} > \frac{1}{n}$, while if $z > \frac{1}{2}$ then $\|z\| = 1 - z > \frac{1}{n}$. This proves $W_+ \subseteq U(n, R)$; the final assertion follows as in Lemma 7.2: on $U(n, R)$ every retained speed exceeds $\frac{1}{n}$, so tightness forces $\min(\|w_1t\|, \|w_2t\|) \leq \frac{1}{n}$ there. $\square$

Lemma 7.9 (Edge reach at $1/q$). *Let $w \geq n$ and suppose that some bad interval of w covers points of W_+ arbitrarily close to $a = \frac{1}{q} + \frac{1}{m_0qn}$. Then $w \geq m_0(n - q) \geq 2(n - q)$.*

Proof. Finitely many closed bad intervals cover a sequence of points of W_+ tending to a from above; one of them, $[\frac{j}{w} - \frac{1}{wn}, \frac{j}{w} + \frac{1}{wn}]$ with centre $c = \frac{j}{w}$, therefore contains a , i.e. $a - \frac{1}{wn} \leq c \leq a + \frac{1}{wn}$. Since $w \geq n > m_0q$, we have $\frac{1}{wn} < \frac{1}{m_0qn}$ and hence

$$c \geq a - \frac{1}{wn} = \frac{1}{q} + \frac{1}{m_0qn} - \frac{1}{wn} > \frac{1}{q},$$

so $d := jq - w$ is a positive integer and

$$\frac{1}{wq} \leq \frac{d}{wq} = c - \frac{1}{q} \leq \frac{1}{m_0qn} + \frac{1}{wn}.$$

Multiplying by wqn gives $n \leq \frac{w}{m_0} + q$, i.e. $w \geq m_0(n - q)$. $\square$

Theorem 7.10 (Two-swaps with a small removal). *Let $q \geq 2$ and $n \geq \max(40, 10q)$, let $R = \{q, r'\}$ with $r' \in [n - 1] \setminus \{q\}$ arbitrary, and let $w_1, w_2 \geq n$. Then $([n - 1] \setminus R) \cup \{w_1, w_2\}$ is not tight.*

Proof. Suppose V tight, and put $\ell^* := \frac{n-q}{qn(n-1)} - \frac{1}{2qn} = \frac{n-2q+1}{2qn(n-1)}$, so that, by $F_{\pm 1} \leq n - 1$ and $m_0 \geq 2$, the total window length $\Lambda := |W_-| + |W_+|$ obeys $\Lambda \geq 2\ell^*$. By Lemma 7.8 the bad sets of w_1, w_2 cover $W_- \cup W_+$, and one bad interval covers points of W_+ arbitrarily close to its inner

edge; by Lemma 7.9 its speed w^* satisfies $w^* \geq 2(n - q)$, while the other speed w' is $\geq n$. By Lemma 7.3,

$$\Lambda \leq \sum_{w \in \{w^*, w'\}} \sum_{J \in \{W_-, W_+\}} \left(w|J| + \frac{2}{n} + 1 \right) \frac{2}{wn} = \frac{4\Lambda}{n} + \frac{4(2/n + 1)}{n} \left(\frac{1}{2(n - q)} + \frac{1}{n} \right),$$

so coverage would force $\ell^*(1 - \frac{4}{n}) \leq \frac{1+2/n}{(n-q)n} + \frac{2(1+2/n)}{n^2}$. We now prove the opposite inequality

$$\ell^* \left(1 - \frac{4}{n} \right) > \frac{1 + 2/n}{(n - q)n} + \frac{2(1 + 2/n)}{n^2} \quad (8)$$

whenever $q \leq \frac{n}{10}$ and $n \geq 40$ (which together imply $4q \leq n - 1$). On the left,

$$\ell^* \left(1 - \frac{4}{n} \right) = \frac{(n - 2q + 1)(n - 4)}{2qn^2(n - 1)} \geq \frac{4}{5} \cdot \frac{9}{10} \cdot \frac{1}{2qn} = \frac{9}{25qn},$$

using $n - 2q + 1 \geq \frac{4n}{5}$ and $\frac{n-4}{n} \geq \frac{9}{10}$; on the right,

$$\frac{1 + 2/n}{(n - q)n} + \frac{2(1 + 2/n)}{n^2} \leq \frac{21}{20} \left(\frac{10}{9n^2} + \frac{2}{n^2} \right) = \frac{49}{15n^2},$$

using $1 + \frac{2}{n} \leq \frac{21}{20}$ and $n - q \geq \frac{9n}{10}$. But $\frac{9}{25qn} > \frac{49}{15n^2}$ is equivalent to $135n > 1225q$, which follows from $q \leq \frac{n}{10}$. This proves (8), the desired contradiction. $\square$

Remark 7.11 (Refined thresholds and the limit constant). After clearing denominators, (8) is equivalent to the positivity of the quartic

$$G(n, q) = n(n - q)(n - 4)(n - 2q + 1) - 2q(n - 1)(n + 2)(3n - 2q),$$

whose leading form in $c = q/n$ is $n^4(6c^2 - 9c + 1)$: the method therefore applies for all $q \leq (\gamma - \varepsilon)n$ once $n \geq n_1(\varepsilon)$, where $\gamma = \frac{9 - \sqrt{57}}{12} = 0.1208\dots$ is the positive root of $6c^2 - 9c + 1$; the clean hypothesis $n \geq 10q$ of Theorem 7.10 is the value $\gamma = \frac{1}{10}$. The exact thresholds $n_0(q) = \min\{n : (8) \text{ holds for all } n' \geq n\}$, computed in exact arithmetic with no later failure (`q_density.py`), are sharper: $n_0(2) = 20$, $n_0(3) = 28$, $n_0(4) = 36$, $n_0(5) = 45$, $n_0(10) = 86$, $n_0(20) = 169$, $n_0(50) = 417$ (asymptotically $n_0(q) \sim 8.3q$). For $q = 2$ the parity refinement $F_{-1} = n - 2$ for odd n (Section 7.1) lowers the threshold to $n \geq 19$ (odd) and $n \geq 20$ (even), recovering Theorem 7.5. The whole chain of this subsection is audited in exact arithmetic by `q_density.py`: the windows of Lemma 7.8 are verified to be *exactly* the two components of $U(n, \{q, r'\})$ adjacent to $\frac{1}{q}$ for all 14,742 instances with $n \leq 60$, $4q \leq n - 1$ and all $r' \neq q$; Lemma 7.9 is verified by brute force over all bad intervals; and (8) is verified on the ranges above.

Corollary 7.12. *For $n \geq 40$, every tight two-swap $([n - 1] \setminus \{r_1, r_2\}) \cup \{w_1, w_2\}$ with $w_i \geq n$ satisfies $\min(r_1, r_2) > n/10$ or $\min(r_1, r_2) = 1$.*

Proof. Apply Theorem 7.10 to $q = \min(r_1, r_2)$ when $2 \leq q \leq n/10$; the hypothesis $n \geq \max(40, 10q)$ holds since $n \geq 40$ and $10q \leq n$. $\square$

7.3 The census

Theorem 7.13 (Two-swap census). *Let $V = ([n-1] \setminus \{r_1, r_2\}) \cup \{w_1, w_2\}$ with $1 \leq r_1 < r_2 \leq n-1$ (any regime, including pairs with both removed speeds beyond $(n-1)/2$) and $n \leq w_1 < w_2 \leq 8n$. An exact-arithmetic census over all $8 \leq n \leq 140$ and all removal pairs finds exactly two tight instances:*

- $n = 8$, $R = \{2, 3\}$, $\{w_1, w_2\} = \{11, 13\}$: $\{1, 4, 5, 6, 7, 11, 13\}$ (Theorem 7.6);
- $n = 74$, $R = \{70, 72\}$, $\{w_1, w_2\} = \{140, 144\}$: the Goddyn–Wong doubling example ([12, Sec. 3]; Corollary 5.2).

Moreover, for $R = \{2, r'\}$ with mid-range r' the exclusion holds unconditionally in the speeds for odd $n \geq 19$ and even $n \geq 20$ (Theorem 7.5), and for $R = \{q, r'\}$ with any removal $2 \leq q \leq n/10$ and any $r' \in [n-1] \setminus \{q\}$ it holds unconditionally for $n \geq \max(40, 10q)$ (Theorem 7.10); by Corollary 7.12, every tight two-swap with $n \geq 40$ and $w_i \geq n$ has both removals in $\{1\} \cup (n/10, n-1]$.

The search bound $w \leq 8n$ is a pragmatic window. It covers the k -GW speeds $2(n-2)$ with a fourfold margin and all sporadic speeds by orders of magnitude; beyond it the census is open, since the effective bound of Section 2 permits speeds up to $O(n^2)$ and, as Question 9.2 makes precise, no unconditional speed bound is known for multi-swaps. The speed-reduction theorem of Tao [24]—to verify the Lonely Runner Conjecture it suffices to check speeds up to $n^{O(n^2)}$ —does not help here: it concerns $\text{LR}(V) \geq 1/n$ for *all* sets, not the classification of the equality case, and likewise the small-velocity lower bounds of [24] exclude tightness only in ranges disjoint from the present search.

The census uses three independent implementations, in complete agreement:

1. *Edge-reach census* (`two_swap_census.py`). For each removal pair, candidate speeds are constrained by necessary edge-reach conditions (an interval of w covering the left end a of a component has an integer j in $[wa - \frac{1}{n}, wa + \frac{1}{n}]$ —at most two candidates per speed and component), surviving pairs are screened in floats with a gap tolerance far below the minimal true gap $\geq 1/(nw_1w_2)$, and every survivor is re-verified in exact rational arithmetic: exact components of $U(n, \{r_1, r_2\})$, exact interval coverage, and $\text{LR}(V) = 1/n$ on the breakpoint set. The screen can only discard non-covering pairs: bad intervals are expanded before floating-point subtraction, so every float component under-approximates the true uncovered region.
2. *Union filter* (`corrected_two_swap_search.py`). For each candidate pair, compute the bad intervals $[j/w_i - 1/(w_in), j/w_i + 1/(w_in)]$, intersect them with the components of $U(n, \{r_1, r_2\})$, and check that the union covers every component.
3. *Remaining-region method* (`two_swap_proof.py`). Compute the *complement*: the set of t where *both* $\|w_1t\| > 1/n$ and $\|w_2t\| > 1/n$, and verify that it is disjoint from $U(n, \{r_1, r_2\})$; this is independent of the component structure.

7.4 The shifted variant

The modifications studied in this paper modify the speed *set*; an orthodox variation modifies the starting line instead. For a vector $\mathbf{v} = (v_1, \dots, v_n)$ of distinct speeds and starting positions

$\mathbf{s} \in [0, 1]^n$, put

$$\gamma(\mathbf{v}; \mathbf{s}) := \max_{t \in [0, 1]} \min_{1 \leq i \leq n} \|s_i + v_i t\|, \quad \gamma^{\min}(\mathbf{v}) := \min_{\mathbf{s} \in [0, 1]^n} \gamma(\mathbf{v}; \mathbf{s}),$$

so that $\gamma(\mathbf{v}; \mathbf{0}) = \text{LR}(\{v_1, \dots, v_n\})$ and $\gamma^{\min}(\mathbf{v})$ is the worst case over starting positions. The *shifted* Lonely Runner Conjecture, suggested by the zonotopal reformulation of Henze and Malikiosis [13], asserts $\gamma^{\min}(\mathbf{v}) \geq \frac{1}{n+1}$ for every $\mathbf{v}$; Malikiosis, Santos and Schymura [18] reduced the classical conjecture to a linearly exponential check and proved the shifted analogue of this reduction conditionally on a two-dimensional assumption. The assumption, and with it the shifted conjecture itself, is false: Blanco, Criado and Santos [5] proved

$$\gamma^{\min}(1, 2, 3, 4, 5) = \frac{15}{94} < \frac{1}{6}, \quad \gamma^{\min}(1, 2, \dots, n) < \frac{1}{n+1} \quad \text{for } 5 \leq n \leq 17,$$

(the case $n = 6$ being $\gamma^{\min}(1, \dots, 6) = \frac{9}{67} < \frac{1}{7}$), while Alcántara, Criado and Santos [1] verified the shifted conjecture for vectors of $n = 4$ speeds and found that among its primitive tight vectors exactly two are tight in the classical sense.

One aspect of the shifted theory is elementary and worth isolating.

Lemma 7.14 (Monotonicity of $\gamma^{\min}$). *If $\mathbf{v}'$ is obtained from $\mathbf{v}$ by deleting one speed, then $\gamma^{\min}(\mathbf{v}') \geq \gamma^{\min}(\mathbf{v})$; equivalently, inserting a speed cannot increase $\gamma^{\min}$. In particular the sequence $\gamma^{\min}(1, 2, \dots, n)$ is nonincreasing in n .*

Proof. Fix starting positions $\mathbf{s}'$ for $\mathbf{v}' = (v_1, \dots, v_{j-1}, v_{j+1}, \dots, v_n)$ and complete them to $\mathbf{s} = (\mathbf{s}', s_j)$ for $\mathbf{v}$ by an arbitrary s_j . For every t ,

$$\min_{i \neq j} \|s_i + v_i t\| \geq \min_{1 \leq i \leq n} \|s_i + v_i t\|,$$

because the minimum is taken over fewer coordinates; maximising over t gives $\gamma(\mathbf{v}'; \mathbf{s}') \geq \gamma(\mathbf{v}; \mathbf{s}) \geq \gamma^{\min}(\mathbf{v})$, and minimising over $\mathbf{s}'$ proves the claim. The last statement follows by deleting the last speed from $(1, \dots, n+1)$. $\square$

The contrast with the classical setting is sharp. Classically, the baseline $[n-1]$ is tight for every n (Dirichlet), the Goddyn–Wong criterion generates infinitely many further tight n -speed sets for every n (Theorems 1.1 and 5.1), and Theorem 1.2 classifies the one-entry modifications exactly. In the shifted setting the baseline itself fails from five speeds on; note that Lemma 7.14 does *not* by itself force the failure to persist (the target $\frac{1}{n+1}$ decreases with n), and indeed each $5 \leq n \leq 17$ is a separate computation of [5]. The same lemma does imply that the shifted extremal quantity $\Gamma(n) := \min_{|\mathbf{v}|=n} \gamma^{\min}(\mathbf{v})$ —the minimum loneliness that must be conceded when the starting positions are chosen adversarially, and the shifted counterpart of the spectrum minimum $\min_{|V|=n} \text{LR}(V)$ —is nonincreasing in n : appending a speed cannot increase $\gamma^{\min}$. Unlike the classical baseline, which attains the minimum $\frac{1}{n}$ for every n , the shifted baseline is not even extremal once it fails: at $n = 6$, $\gamma^{\min}(2, 3, 4, 5, 6, 8) = \frac{2}{15} < \gamma^{\min}(1, 2, \dots, 6) = \frac{9}{67}$ [5]. Determining $\Gamma(n)$ and its minimisers, and deciding whether the shifted conjecture fails for every $n \geq 5$, are natural open problems (Question 9.4); among all vectors of five speeds with $\sum v_i \leq 100$, exhaustive computation in [5] suggests that $(1, 2, 3, 4, 5)$ is the only counterexample.

8 Verification protocol

All computations use exact rational arithmetic (`fractions.Fraction`). Each verification below is a *mathematical proof by finite enumeration*, not a numerical experiment: the breakpoint principle reduces $\text{LR}(V)$ to evaluation on a finite set $T(V)$, and the search ranges are reduced to finite size by the effective bound ($w \leq 4rI/(2s - I)$, Section 2), by the Kanold bound ($s \leq 64$, Remark above), or by the edge-reach masks of Section 7. The floating-point pre-pass is a convenience only; every reported value is certified in exact arithmetic.

8.1 Searches

verification	range	size	outcome
mid-range classification	$r \leq 30, n \in [2r + 1, 6r],$ $w \leq 4rI/(2s - I)$	21,789	$r \geq 3$: none tight; $r = 2$: exactly (5, 7), (6, 9)
brute scan, no eff. bound	$r \leq 8, w \leq 30n$		none tight
all-components filter	$r \leq 16$		no survivors ($r \geq 3$); $r = 2$: exactly (5, 7), (6, 9)
$M(n)$ from (1)	$n \leq 40000$		$< 0.60n \log n + 52n$ always
primorial family	$r \leq 9699690$		w/n up to 11.00; $m = (q - 1)/2$ throughout
<i>New computations (this paper):</i>			
density chain (Lemmas 7.1–7.4)	$n \leq 199$	—	inequality holds for every odd $n \geq 19$ /even $n \geq 20$
general- q chain (Sec. 7.2)	$n \leq 60, 4q \leq n - 1, \text{ all } r'$ $q \leq 100, n \leq 5000$	14,742 exact scan	windows exact; (8) holds for $n \geq \max(40, 10q)$
deletion values (Thm. 2.8, Prop. 2.9 audit)	$n \leq 30, \text{ all } r$	432	closed form for $r = 2$; $\text{LR} \geq \frac{1}{n-1}$, equality iff $r = n - 1$
two-swap census, all pairs	$n = 8 \dots 140, w \leq 8n$	447,545 triples 98,463 pairs, 2 exact	only $n = 8$ sporadic + GW doubling $n = 74$ tight
two-swap $\{2, r'\}$, deep window	$n = 9 \dots 18, w \leq 40n$	96,725 pairs	none tight
GW doubling $n = 74$	exact breakpoint	1	tight ($\text{LR} = 1/74$)
GW doubling $n = 284$	branch-and-bound	1	tight (60,979 cells)
$k = 3, r = 3 \dots 15$	$n = 2r+1 \dots 4r$	200,661	0 tight
$k = 3, r = 16 \dots 35$	$n = 2r+1 \dots 4r$	41,038	0 tight
$k = 4, n = 10 \dots 30$	all multipliers	12,646	0 tight
<i>Total: 820,404 enumerated cases, 0 unexpected tight sets.</i>			

The instance $r = 210, n = 212, w = 1050$ (the extremal-ratio record) was additionally certified tight by an adaptive branch-and-bound procedure that does not enumerate breakpoints, in 33,665 cells. The two-swap census was cross-checked by the two legacy searches (union filter over $n = 8 \dots 17, w \leq 10n$, and remaining-region method over $n = 8 \dots 17, w \leq 8n$, plus the $\{2, 3\}$ scan to $n = 24, w \leq 6n$), with complete agreement; per- n case counts are recorded in `two_swap_census.log`.

8.2 Spectrum values near the minimum

Proposition 2.6 places every single-speed modification value in $[1/n, \text{LR}([n - 1] \setminus \{r\})]$. For $r = 2$ the upper bound is the closed form of Theorem 2.8, $\text{LR}([n - 1] \setminus \{2\}) = 2/(2\lfloor n/2 \rfloor + 3)$,

independently confirmed in exact arithmetic for $5 \leq n \leq 30$ (`q_density.py`); it is strictly above $1/(n-1)$, so for $r = 2$ the sandwich interval contains the entire “isolated region” $(1/n, 1/(n-1))$ between the tight value and the first accumulation point of $\tilde{S}(n-1)$.

For each $n \leq 12$ with $r = 2$, the smallest non-tight value $\text{LR}([n-1]_{2 \rightarrow w})$ falls *inside* this isolated region for $n \leq 8$ and matches the pattern $s/((n-1)s+1)$ predicted by [11, Conj. 1.1]—a pattern that Fan and Sun [10] showed cannot persist in general, disproving the spectrum conjecture of Kravitz at four speeds by an infinite family of counterexamples; for $n \geq 9$ the smallest value jumps *past* the accumulation point $1/(n-1)$. The transition is explained by the effective bound: for odd n the bound gives $w \leq 8$, so already $w \geq n \geq 9$ exceeds it, and the component of $U(n, 2)$ is too long for w to cover. In spectral terms, the isolated region $(1/n, 1/(n-1))$ predicted by the well-ordering of $\tilde{S}(n-1)$ is populated by modification values only while modifiers exist below the effective bound; once the bound closes the window, modification values skip directly past the first accumulation point, so the isolation visible in the table is a finite- n shadow of the spectral structure.

n	$1/n$	next value	w	in $(\frac{1}{n}, \frac{1}{n-1})$?	matches $\frac{s}{(n-1)s+1}$
5	1/5	2/9	5	yes	$s=2$
6	1/6	2/11	7	yes	$s=2$
7	1/7	2/13	9	yes	$s=2$
8	1/8	2/15	11	yes	$s=2$
9	1/9	2/15	11	no ($> 1/8$)	—
10	1/10	2/17	13	no ($> 1/9$)	—
11	1/11	2/17	13	no ($> 1/10$)	—
12	1/12	2/19	15	no ($> 1/11$)	—

8.3 Audit of the proofs

Beyond the searches, each inequality used in Section 3 was tested directly as an independent exact-arithmetic check (the theoretical proof of Lemma 3.3 covers all $r \geq 3$ unconditionally): the equivalence of Lemma 3.1 with the exact integer-existence condition, and the sign claims of Lemma 3.2, over all 2628 admissible triples (n, r, w) with $3 \leq r \leq 14$ (no discrepancy, and no triple satisfied both windows); the implication $\varphi(r) \geq 4 \Rightarrow 2s - I \geq 5$ for all $r \leq 60$ (no violation; in fact $2s - I \leq 4$ occurred for $r = 3$ only among $r \geq 3$; $r = 2$ also gives small values at $(5, 2)$, $(6, 2)$, but $\varphi(2) = 1 < 4$ so the implication is vacuous there). This implication is the finite-range shadow of the B4 argument: $d = \delta < j(r)$ keeps $2s - I = s - d$ away from 0, and it served as the gateway in an earlier computational version of the proof. The threshold chain $2s - I \geq 5 \Rightarrow \frac{4rI}{2s-I} < r(n-2)$ for all $r \leq 60$ (exact, no violation); the key inequality $4rI/(2s - I) < r(n-2)$ itself for all $r \leq 60$ and all admissible n (confirming Lemma 3.3); and the Rosser–Schoenfeld constants numerically for $x \leq 2 \cdot 10^6$ using exact integer summation ($\min_{x \geq 101} \theta(x)/x = 0.84976$ on the range).

Remark 8.1 (Spectrum conjectures). The pattern $s/(ms+1)$ deserves care. Conjecture 1.1 of Giri–Kravitz [11] predicts that every value of $\tilde{S}(m)$ is either $\geq \frac{1}{m}$ or of the form $s/(ms+1)$, $s \geq 1$; the table above tests its immediate-successor case $s = 2$ on the class of one-entry modifications of the baseline, where it holds for $n \leq 8$. For general four-speed sets, however, Fan and Sun [10] constructed an infinite family of counterexamples to this conjecture and proposed an amended version, which they confirm for four-speed sets containing a pair of speeds with a common factor

≥ 3 . Our verification of the $s = 2$ pattern and their amended conjecture address disjoint classes and neither implies the other.

9 Open problems

Question 9.1. Classify the tight *two-entry* modifications $([n - 1] \setminus \{r_1, r_2\}) \cup \{w_1, w_2\}$ for arbitrary speeds. The census (Theorem 7.13) settles $n \leq 140$, $w \leq 8n$; removals $\{2, r'\}$ with mid-range r' are settled unconditionally in the speeds for $n \geq 19$ (odd)/ $n \geq 20$ (even) by the density mechanism of Section 7.1, and removals $\{q, r'\}$ with any $2 \leq q \leq n/10$ and any $r' \neq q$ for $n \geq \max(40, 10q)$ by its generalisation (Theorem 7.10); by Corollary 7.12 the residual open regime is *both* removals in $(n/10, n - 1]$, or a removal equal to 1. Sharpening the constant $1/10$ towards the limit $\gamma = (9 - \sqrt{57})/12$ of the method (Remark 7.11) is bookkeeping. In the both-large subregime $r_1, r_2 > (n - 1)/2$, the extended divisibility lemma (Lemma 6.1) forces each inserted speed covering $1/r_i$ to be a multiple of r_i , so the Goddyn–Wong criterion governs each swap separately and only their interaction remains; the census suggests the sharp statement: *every tight two-swap with $w_i \geq n$ and $r_1, r_2 > (n - 1)/2$ arises from the multi-acceleration Theorem 5.1, i.e. $w_i = m_i r_i$ with (4).* For the double mid-range case, whether the window mechanism of Section 3, applied to each removed speed separately, can reduce it to a bounded residual remains open.

Question 9.2. For one-entry modifications, Theorem 1.3 gives $\max V \leq 0.60 n \log n + 52 n$ with sharp constant $\frac{1}{2}$ (attained along $n = p\# + 2$), and Theorem 6.2 extends the upper bound to all families obtained from the multi-acceleration Theorem 5.1. Does the same bound, or any sub-quadratic bound, hold for general multi-swap modifications $V = ([n - 1] \setminus R) \cup W$? Two obstructions are known. For mid-range $r \leq (n - 1)/2$, Lemma 6.1 may be witnessed by a retained speed, leaving inserted speeds $w \in W$ unconstrained by divisibility. Even in the GW regime ($r > (n - 1)/2$), for $k \geq 2$ the Goddyn–Wong criterion (1) does not follow from the tightness of V , so the Chebyshev argument of Theorem 1.3 does not apply per speed. In both cases the coverage of the joint uncovered region $U(n, R) \supseteq U(n, r)$ by W is a union condition that can split within components (Section 7), so the interval lemma does not directly yield the single-swap bound. No counterexample was found in the exact-arithmetic searches of Section 8 ($k \leq 3$, $n \leq 140$; $k = 4$, $n \leq 30$).

Question 9.3. Classify tight k -swap modifications for $k \geq 2$. Theorem 5.1 gives a general sufficient condition, and its converse already fails in general [12, Sec. 3]; are there tight multi-swaps that do not arise from it? Exact arithmetic verification ($k = 3$, $n \leq 140$; $k = 4$, $n \leq 30$; 820,404 enumerated cases in total, Section 8) found none beyond the Goddyn–Wong families and the sporadic $\{1, 4, 5, 6, 7, 11, 13\}$.

Question 9.4 (Shifted analogue). In the notation of Section 7.4: is $\Gamma(n)$, the minimum of $\gamma^{\min}(\mathbf{v})$ over n -speed vectors, strictly below $\frac{1}{n+1}$ for every $n \geq 5$, as computed for $5 \leq n \leq 17$ in [5]? What are the minimisers of $\Gamma(n)$ —the shifted analogue of the tight instances, for which no Goddyn–Wong-type supply is known—and how do they relate to the classical minimisers $[n - 1]_{r \rightarrow w}$ of Theorem 1.2? At five speeds the two notions already diverge: among the primitive shifted-tight vectors found in [1], exactly two are tight in the classical sense.

Data availability

The verifications of Section 8 are collected in the repository

<https://github.com/Qinrayn/lonely-runner-tight-II>

short standalone programs in exact rational arithmetic (Python, `fractions.Fraction`), one per verification claim, together with the run logs (e.g. `two_swap_census.log`).

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